



Patient-Aware Explainable Machine Learning for Preoperative Classification of Benign and Malignant Breast Lesions Using Single- and Multiparametric MRI Radiomics

Elmira Yazdani^{1*}, Mahla Entezari², Zahra Farzanehgan¹, Hengameh Nazari³; Elahe Mirzaee⁴, Zahra Bagherpour¹, Pedram Fadavi^{4,5}, Saeed Hosseini Toudeshki⁶, Hasan Abdollahzadeh⁷, Mojtaba Safari⁸, Saeed Reza Kheradpisheh^{2*}, Manijeh Beigi^{4,5*}

¹*Department of Medical Physics, School of Medicine, Iran University of Medical Sciences, Tehran, Iran*

²*Department of Computer and Data Sciences, Faculty of Mathematical Sciences, Shahid Beheshti University, Tehran, Iran*

³*Department of Radiology and Nuclear Medicine, School of Medicine, Mazandaran University of Medical Sciences, Mazandaran, Iran*

⁴*Department of Radiation Oncology, School of Medicine, Iran University of Medical Sciences, Tehran, Iran*

⁵*Breast Health & Cancer Research Center, Iran University of Medical Sciences, Tehran, Iran*

⁶*Department of Radiation Oncology, Sepid Nikan Hospital, Arina Imaging Center, Tehran, Iran*

⁷*School of Medicine, Tehran University of Medical Sciences, Tehran, Iran*

⁸*Department of Radiation and Cellular Oncology, The University of Chicago, IL, 60637, USA*

Corresponding Authors:

- Dr. Manijeh Beigi

Email: beiki.m@iums.ac.ir; Beigi310@gmail.com

Assistant Professor at the Department of Radiation Oncology, School of Medicine, Iran University of Medical Sciences, Tehran, Iran

- Dr. Saeed Reza Kheradpisheh

Email: s_kheradpisheh@sbu.ac.ir; sr.kheradpisheh@gmail.com

Assistant Professor at the Department of Computer and Data Sciences, Faculty of Mathematical Sciences, Shahid Beheshti University, Tehran, Iran

- Dr. Elmira Yazdani

Assistant Professor at the Department of Medical Physics, School of Medicine, Iran University of Medical Sciences, Tehran, Iran

Email: Yazdani.el@iums.ac.ir; lmirayazdani@gmail.com

ABSTRACT

Background: Breast magnetic resonance imaging (MRI) provides high sensitivity for breast lesion detection; however, differentiating benign and malignant lesions remains challenging due to overlapping imaging characteristics. This study aimed to develop an explainable, patient-aware machine learning (ML) framework using single- and multiparametric MRI radiomics for classification of benign and suspicious breast lesions based on Breast Imaging Reporting and Data System (BI-RADS) assessment.

Methods: In this retrospective study, MRI data from 49 patients were analyzed. Lesions were classified according to the BI-RADS categories, with BI-RADS 1–3 considered benign and BI-RADS 4–5 considered malignant. Multiparametric MRI inputs included ADC maps, T2-weighted (T2W) images, and dynamic contrast-enhanced (DCE) T1-weighted (T1W) images (pre-contrast, post1- and post2-contrast phases). Lesions were manually segmented on post1 DCE-T1W images by an experienced breast radiologist. After image preprocessing, radiomic features were extracted from five MRI sources. A total of 31 single- and multi-sequence feature configurations were evaluated using a patient-aware nested cross-validation (CV) framework. Feature selection strategies, ML classifiers, and Synthetic Minority Oversampling Technique (SMOTE)-based class balancing were systematically compared. SHapley Additive exPlanations (SHAP) analysis was applied for model interpretation.

Results: The best-performing model was obtained from the ADC+DCE-Post1+DCE-Pre feature configuration using Elastic Net followed by Minimum Redundancy Maximum Relevance (mRMR) feature selection, automatic feature-number optimization, and a linear SVM classifier without SMOTE. This model achieved a mean outer-fold ROC-AUC of 0.836 ± 0.088 and a **pooled ROC-AUC of 0.8**. Increasing the number of MRI sources did not consistently improve performance. SHAP analysis demonstrated that ADC-derived first-order features and DCE-derived intensity and morphological features were the most influential predictors.

Conclusion: A patient-aware explainable ML framework based on multiparametric MRI radiomics can provide quantitative discrimination of benign and malignant breast lesions. The findings highlight the importance of robust validation and selective feature fusion rather than simple accumulation of imaging sources. Further multicenter validation is required before clinical translation.

Keywords: Breast MRI; Radiomics; Machine Learning; Explainable AI; ADC; DCE-MRI; Breast Lesion Classification

1. BACKGROUND

Breast cancer represents the most commonly diagnosed malignancy among women worldwide and remains a major public health challenge [1]. Imaging evaluation plays a fundamental role in the detection and characterization of suspicious breast lesions, with the Breast Imaging Reporting and Data System (BI-RADS) providing a standardized framework for imaging interpretation, reporting, and clinical management [2].

Breast magnetic resonance imaging (MRI) has emerged as one of the most sensitive imaging techniques for breast cancer evaluation, particularly in cases with inconclusive findings from conventional imaging modalities [3, 4]. Breast MRI has become an important complementary imaging modality in the evaluation of suspected breast cancer and the screening of high-risk populations. However, breast MRI is a complex examination that generates large volumes of multiparametric data from different imaging sequences and contrast phases. Accurate interpretation therefore requires a structured approach to integrate diverse imaging characteristics, including lesion morphology, enhancement kinetics, and tissue properties [5]. In the absence of objective and standardized decision-support tools, the complexity of MRI interpretation and its dependency on interpretation of radiologist expertise may contribute to false-positive assessments and additional unnecessary diagnostic procedures [6]. The evaluation of multiple imaging features, including lesion morphology, enhancement kinetics, and tissue characteristics, may introduce variability between observers. Furthermore, although BI-RADS provides standardized terminology, the final assessment still relies on the integration of multiple imaging findings by the interpreting radiologist and does not provide a fully objective quantitative decision-making framework.

Machine learning (ML) has emerged as a promising approach to address these limitations by enabling automated analysis of complex imaging patterns that may not be readily recognized through conventional visual assessment [7-9]. By learning relationships between quantitative imaging characteristics, ML algorithms can identify subtle associations among lesion features and generate predictive models for distinguishing benign and malignant breast lesions. Previous studies have demonstrated the potential of ML-based approaches in breast MRI analysis, showing that quantitative models can provide objective malignancy probability estimates and serve as complementary decision-support tools for suspicious breast findings. Rather than replacing radiologist interpretation, these approaches may improve diagnostic confidence by integrating complex imaging information into reproducible quantitative predictions [10-12].

Radiomic analysis is an advanced image-processing approach that converts medical images into quantitative data by applying mathematical and computational methods [13-15]. These extracted features provide information about imaging characteristics that may not be readily detectable through visual assessment alone. Radiomic descriptors commonly include measurements of lesion shape, voxel intensity distribution, texture patterns, and statistical properties, which may reflect underlying tissue composition and biological heterogeneity. In breast imaging, malignant lesions often demonstrate irregular morphology, heterogeneous internal architecture, and complex enhancement behavior compared with benign findings. Radiomic features enable quantitative characterization of these differences and provide potential imaging biomarkers for ML-based lesion classification [16, 17].

Despite the potential of radiomics and machine learning for breast lesion classification, developing reliable predictive models remains challenging. One important limitation is the potential presence of class imbalance,

where the number of benign and malignant cases may not be equally represented within a dataset. Such imbalance can bias ML algorithms toward the majority class and reduce their ability to accurately identify minority-class samples, which are often the clinically important cases. To mitigate this issue, different strategies have been proposed, including data-level approaches such as oversampling of minority samples or undersampling of majority samples, as well as algorithm-level approaches that incorporate class weighting during model training. However, the effectiveness of these methods depends on dataset characteristics and should be carefully evaluated to avoid introducing artificial patterns or reducing model generalizability.

Another important consideration in breast MRI radiomics is the integration of multiparametric imaging information. Breast MRI provides complementary biological information through different sequences, including diffusion-weighted imaging (DWI)/apparent diffusion coefficient (ADC), T2-weighted (T2W) imaging, and dynamic contrast-enhanced (DCE) MRI. ADC-derived features are associated with tissue diffusivity and cellular density, whereas DCE-MRI features reflect vascular permeability, enhancement kinetics, and lesion heterogeneity. Although combining multiple MRI sequences may theoretically improve lesion characterization, increasing the number of imaging sources also increases feature dimensionality and the possibility of including redundant or unstable predictors. Therefore, the optimal combination of MRI sequences remains unclear, and simply incorporating additional imaging information does not necessarily guarantee improved classification performance.

Furthermore, reliable evaluation of radiomics-based ML models requires rigorous validation strategies. Many previous studies have been conducted using random splitting of lesions or regions of interest (ROI), which may lead to data leakage when multiple lesions originate from the same patient. In such cases, correlated samples from the same individual may appear in both training and testing datasets, resulting in overly optimistic performance estimates. Patient-level separation during feature selection, model optimization, and evaluation is therefore essential to provide a realistic estimation of model generalizability. In addition, limited interpretability remains a barrier to clinical adoption of artificial intelligence (AI) models. Explainable artificial intelligence (XAI) approaches provide insight into the contribution of individual features and may improve the transparency and clinical acceptance of machine-learning systems.

Although previous studies have demonstrated the feasibility of MRI radiomics for breast lesion classification, several limitations remain [10, 12, 17-19]. Many investigations have focused on a single imaging sequence, evaluated a limited number of classifiers, or applied feature selection and validation strategies without comprehensive comparison. In particular, the combined effects of MRI source selection, feature-selection strategy, class-balancing method, and classifier choice have not been systematically investigated within a patient-aware validation framework.

Therefore, the aim of this study was to develop and systematically evaluate an explainable patient-aware ML framework for differentiating benign and malignant breast lesions using multiparametric MRI radiomics. Patient-aware refers to maintaining all lesions from the same patient within the same cross-validation partition. Five MRI sources, including ADC maps, pre-contrast T1-weighted images, two post-contrast DCE phases (Post1 and Post2), and T2-weighted images, were investigated across all possible non-empty combinations. The impact of different feature-selection strategies, class-balancing approaches, and ML classifiers was assessed using nested patient-

aware cross-validation (CV) to minimize overfitting and prevent data leakage. Finally, XAI analysis was applied to identify the most influential radiomic predictors and evaluate their biological relevance.

2. MATERIALS AND METHODS

This study developed and evaluated an explainable ML framework to classify benign and malignant breast lesions using multiparametric MRI radiomic features. The overall workflow consisted of nine sequential stages, beginning with patient enrollment and MRI data acquisition and progressing through lesion segmentation, image preprocessing, radiomic feature extraction, feature selection, machine-learning model development, model evaluation, and clinical interpretation (Figure 1).

In the first stage (Patient Enrollment), we included patients with breast lesions diagnosed on MRI and collected relevant clinical information, imaging characteristics, pathological outcomes, and radiological assessments. Lesions were categorized according to BI-RADS classification, with BI-RADS 1–3 considered benign and BI-RADS 4–5 considered suspicious/malignant according to radiological evaluation and available reference standards.

In the second stage (MRI Data Acquisition), multiparametric breast MRI examinations were analyzed, including pre-contrast T1W imaging, DCE-T1W, T2W, and ADC maps derived from DWI. These complementary MRI sequences provided information regarding lesion morphology, vascularity, tissue composition, and cellular characteristics.

The third stage (Lesion Segmentation) involved delineation of breast lesions on MRI images. Lesions were independently segmented by an experienced radiologist to generate segmentation masks. These masks were subsequently used to define the ROIs for radiomic analysis.

In the fourth stage (Image Preprocessing), standardized preprocessing procedures were applied to improve radiomic feature reproducibility and reduce variability between MRI examinations. These procedures included image registration between MRI sequences, voxel resampling, intensity normalization, and gray-level discretization prior to feature extraction.

In the fifth stage (Radiomic Feature Extraction), quantitative imaging features were extracted from the segmented lesion regions across different MRI sequences. Extracted features included first-order intensity statistics, shape-based features, and texture features describing intratumoral heterogeneity. Radiomic feature extraction was conducted separately for individual MRI sequences and different segmentation masks to investigate the impact of imaging modality and segmentation variability.

In the sixth stage (Radiomic Feature Selection), redundant and non-informative features were removed to reduce feature dimensionality and improve model robustness. Feature filtering, redundancy reduction, and selection of informative predictive features were conducted to identify the most relevant radiomic features associated with the classification of benign and suspicious breast lesions based on BI-RADS assessment.

In the seventh stage (ML Model Development), supervised ML algorithms were trained using the selected radiomic features. Models were developed using individual MRI datasets as well as combined multiparametric

feature sets. Different MRI combinations, including single-sequence models, pairwise combinations (**T1+DCE/T2+ADC**), and combined multiparametric models (T1+T2+ADC), were evaluated. Appropriate validation strategies, including CV approaches suitable for small medical imaging datasets, were applied to minimize overfitting.

In the eighth stage (ML Model Evaluation), the performance and generalizability of developed models were assessed using multiple evaluation metrics, including receiver operating characteristic (ROC) analysis, area under the curve (AUC), accuracy, precision, recall, confusion matrices, predictive values, Cohen's kappa, and calibration analysis. Statistical comparisons between models were conducted with appropriate correction for multiple testing.

In the ninth stage (Clinical Interpretation and Clinical Impact Assessment), XAI approaches were applied to interpret model predictions and identify the most influential radiomic features contributing to classification decisions. SHapley Additive exPlanations (SHAP) analyses were used to provide feature-level interpretability and facilitate clinical understanding of model behavior. The following sections provide detailed descriptions of each component of the proposed workflow.

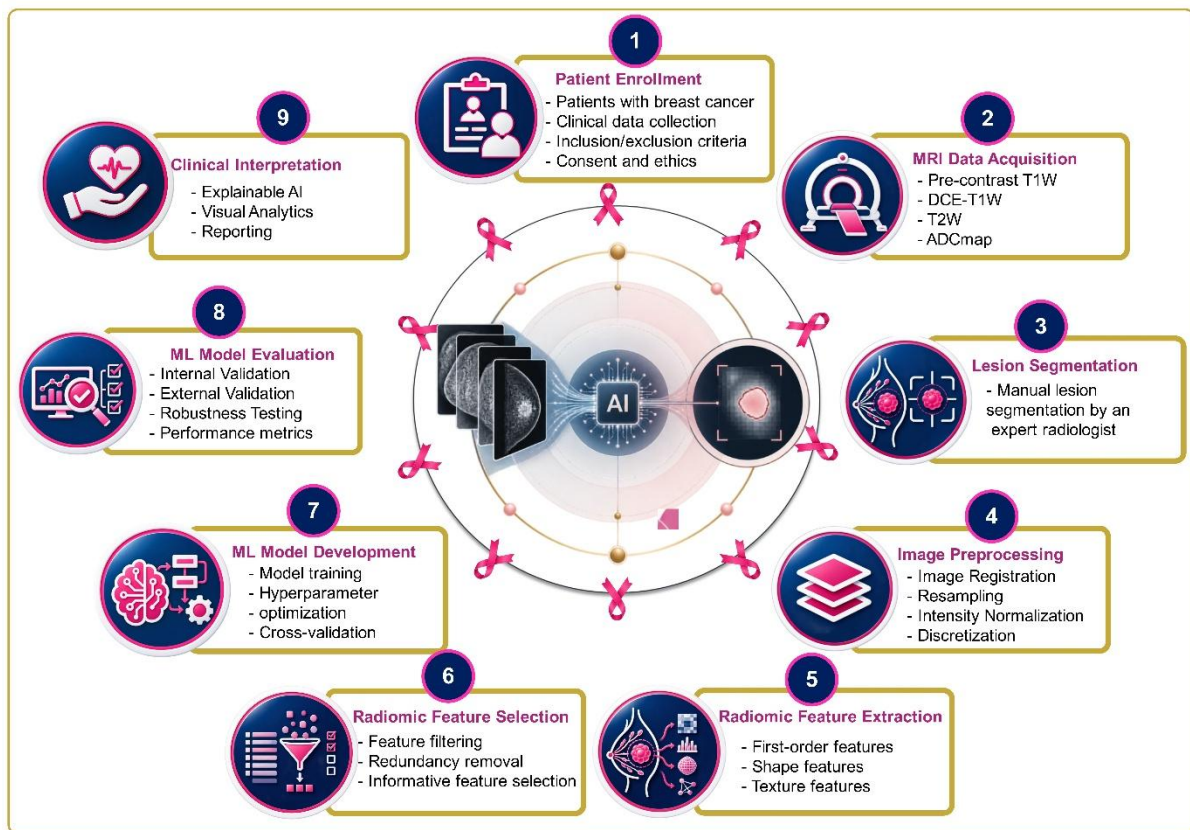


Figure 1. Overall workflow of the proposed patient-aware explainable machine-learning framework for benign and malignant breast lesion classification using multiparametric MRI radiomics. The pipeline includes patient enrollment, MRI acquisition, lesion segmentation, image preprocessing, radiomic feature extraction and selection, machine-learning model development, validation, and explainable AI-based clinical interpretation. patient-level separation was maintained throughout preprocessing, feature selection, and validation.

2.1. Patient Enrollment

This retrospective study included female patients who underwent general breast MRI as part of the clinical evaluation of breast abnormalities. Breast MRI was conducted to further characterize imaging findings, identify and localize breast lesions, assess their degree of suspicion, and determine whether additional diagnostic investigation, particularly biopsy, was warranted. Only patients with an available radiologist-assigned BI-RADS assessment and adequate multiparametric MRI data were considered eligible for subsequent radiomic analysis.

All MRI examinations were reviewed by experienced breast radiologists. The right and left breasts were evaluated separately, and detected abnormalities were localized and assigned a BI-RADS category according to their imaging characteristics. BI-RADS categories reflected the radiologist's level of suspicion and guided subsequent clinical management. BI-RADS 1 represented a negative examination, BI-RADS 2 a benign finding, and BI-RADS 3 a probably benign finding. BI-RADS 4 represented a suspicious abnormality for which biopsy was recommended, whereas BI-RADS 5 indicated a finding highly suggestive of malignancy for which biopsy was strongly recommended.

For ML analysis, the radiologist-assigned BI-RADS categories were converted into a binary classification endpoint. BI-RADS categories 1–3 were assigned label 0, representing the benign/low-suspicion class, whereas BI-RADS categories 4–5 were assigned label 1, representing the malignant/high-suspicion class. The binary labels therefore represented the radiological level of suspicion derived from the BI-RADS assessment. BI-RADS assessments for the right and left breasts were recorded separately and linked to the corresponding lesion location to ensure appropriate lesion-level labeling.

Demographic information, including patient age and body weight, was obtained from the available clinical records. The patients had a mean age of 43.8 years (range, 23–72 years) and a mean body weight of 67.2 kg (range, 43–85 kg).

Patients were eligible for inclusion when they met the following criteria: (1) availability of a multiparametric breast MRI examination containing the sequences required for radiomic analysis, including DCE-T1W, T2W, and DWI with derived ADC maps; (2) presence of an identifiable breast abnormality or lesion evaluated by a radiologist; (3) availability of a documented BI-RADS assessment for the corresponding breast; and (4) sufficient image quality for lesion identification, segmentation, preprocessing, and subsequent radiomic feature extraction. Cases were excluded when BI-RADS information was unavailable, the required MRI sequences were incomplete, severe imaging artifacts compromised lesion assessment, or the target lesion could not be reliably identified or segmented.

The study protocol was reviewed and approved by the local Institutional Review Board/Ethics Committee of our institution. All procedures were conducted in accordance with the ethical principles of the Declaration of Helsinki and its subsequent amendments. Given the retrospective nature of the study and the use of previously acquired clinical imaging and medical-record data, the requirement for written informed consent was waived by the Ethics Committee. All patient data were anonymized before image processing and analysis to protect patient confidentiality.

2.2. MRI Data Acquisition

All breast MRI examinations were acquired on a 1.5 T Siemens MRI system (Siemens Healthineers, Germany) with patients positioned head-first prone. The multiparametric MRI protocol comprised T2-weighted short-tau inversion recovery imaging (T2W-STIR), DWI with derived ADC maps, and DCE-T1W. Representative images from the MRI protocol are shown in Figure 2, and sequence-specific acquisition parameters are summarized in Table 1. The examinations were conducted between June 2022 and February 2026.

T2W images were acquired in the transverse plane using a two-dimensional turbo spin-echo inversion-recovery sequence (T2-TSE STIR). DWIs were acquired using a transverse echo-planar imaging (EPI) sequence with b-values of 50 and 800 s/mm², from which ADC maps were generated by the scanner software. DWI was not independently included in radiomic feature extraction because the ADC map provides a quantitative representation of tissue diffusivity derived from the diffusion-weighted acquisition. Including both DWI and ADC would introduce partially redundant diffusion-related information while substantially increasing the feature dimensionality in this relatively small cohort. Therefore, we used the quantitatively derived ADC map as the diffusion-sensitive input for radiomic analysis, while retaining DWI only as the source sequence for ADC generation.

DCE-MRI was acquired using a three-dimensional fat-suppressed T1-weighted gradient-echo sequence (3D VIBE/DynaVIEWS with SPAIR fat suppression). The dynamic acquisition consisted of one pre-contrast phase followed by six post-contrast phases after intravenous administration of a gadolinium-based contrast agent at an injection rate of 2 mL/s. For the present radiomic analysis, the pre-contrast phase and the first two post-contrast phases (DCE-Pre, DCE-Post1, and DCE-Post2) were retained, whereas the remaining post-contrast phases were not included. The second post-contrast phase was used as the reference image for lesion delineation because of its high lesion conspicuity.

Accordingly, five MRI image sets per lesion were included in the subsequent radiomics workflow: DCE-Pre, DCE-Post1, DCE-Post2, T2W-STIR, and ADC. Because the native spatial resolution differed among sequences, spatial harmonization was subsequently conducted during image preprocessing before radiomic feature extraction.

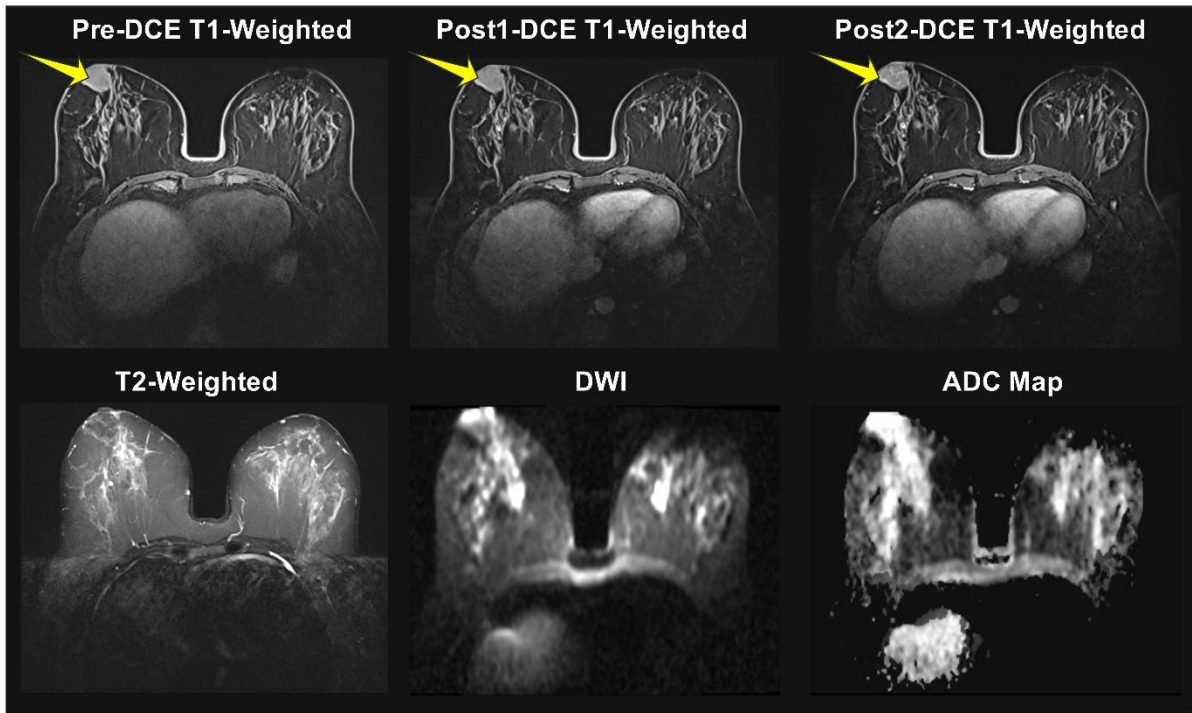


Figure 2. Representative multiparametric breast magnetic resonance imaging (MRI) images used in the study, including dynamic contrast-enhanced T1-weighted imaging (DCE; pre-contrast, first post-contrast, and second post-contrast phases), T2-weighted short-tau inversion recovery imaging (T2W-STIR), diffusion-weighted imaging (DWI), and the corresponding apparent diffusion coefficient (ADC) map. DWI is shown as the source sequence for ADC generation but was not independently included in radiomic feature extraction.

Table 1. MRI acquisition parameters for the sequences included in radiomic analysis.

Parameter	T2W-STIR	DWI/ADC	DCE-T1W
Acquisition technique	2D TSE-STIR	2D EPI	3D GRE/VIBE
Protocol/sequence	t2_tse_stir_SG	ep2d_diff_tra_spair	t1_fl3d_tra_dynaVIEWS_spairM
TR (ms)	7110	7100	4.9
TE (ms)	59	100	2.39
TI (ms)	170	–	–
Flip angle (°)	170	90	10
Slice thickness (mm)	4	4	1.6
Slice spacing (mm)	5.0	5.2	1.6*
Volume dimensions	432 × 384 × 30	344 × 184 × 30	372 × 384 × 104
Voxel spacing (mm ³)	0.781 × 0.781 × 5.0	1.163 × 1.163 × 5.2	0.940 × 0.940 × 1.6
Number of averages	1	6	1
Echo train length	11	88	1
Pixel bandwidth (Hz/pixel)	200	1265	465
b-values (s/mm ²)	–	50, 800	–
Fat suppression	STIR	SPAIR	SPAIR

* For DCE-T1W, 1.6 mm corresponds to the through-plane spacing observed in the reconstructed Slicer volume; the DICOM slice thickness was also approximately 1.6 mm.

Abbreviations: ADC, apparent diffusion coefficient; DCE, dynamic contrast-enhanced; DWI, diffusion-weighted imaging; EPI, echo-planar imaging; GRE, gradient echo; MRI, magnetic resonance imaging; SPAIR, spectral attenuated inversion recovery; STIR, short tau inversion recovery; T1W, T1-weighted imaging; T2W, T2-weighted imaging; TE, echo time; TI, inversion time; TR, repetition time; TSE, turbo spin echo; VIBE, volumetric interpolated breath-hold examination.

2.3. Lesion Segmentation

Lesion segmentation was conducted on the selected post-contrast DCE-T1W images, which demonstrated optimal lesion conspicuity and contrast differentiation between abnormal tissue and surrounding breast parenchyma. As illustrated by the yellow arrows in Figure 2, the target breast lesions were identified using the pre- and post-contrast DCE-T1W images and subsequently delineated for radiomic analysis. All target lesions were manually delineated by an experienced breast radiologist with 7 years of experience in breast MRI interpretation.

Manual segmentation was conducted using ITK-SNAP software (version 2.2.2) [20]. The radiologist reviewed the DCE-T1W images acquired in the prone position and manually generated three-dimensional ROIs encompassing the entire lesion volume. The resulting segmentation masks were exported in NIFTI format and subsequently used for radiomic analysis.

Because breast MRI datasets were acquired in the prone position, and the exported ITK-SNAP segmentation masks did not always preserve the original image spatial geometry and orientation information, an additional mask-to-image alignment step was applied before radiomic feature extraction. A custom alignment algorithm developed using the SimpleITK Python library ensured accurate voxel-wise correspondence between the manually generated masks and the original DCE-T1W images. The algorithm corrected orientation discrepancies, matched the mask geometry to the reference MRI volume, and transferred the original image spatial information, including voxel spacing, origin, and direction matrix, to the segmentation masks.

The corrected segmentation masks were then visually inspected to confirm accurate overlap with the corresponding DCE-T1W images. Only geometrically aligned masks were used in subsequent preprocessing, radiomic feature extraction, and ML analyses.

2.4. Image Preprocessing

Prior to radiomic feature extraction, all MRI datasets underwent standardized preprocessing to reduce variability related to differences in image intensity distribution and spatial resolution. The preprocessing pipeline was applied to the five MRI inputs used in the radiomic workflow: ADC, DCE-Pre, DCE-Post1, DCE-Post2, and T2W, using LIFEx software V25.06.1 [21]. To improve workflow efficiency and ensure consistent processing across all patients and MRI sequences, a custom automation script was developed to facilitate batch execution of the LIFEx feature extraction workflow and reduce manual intervention.

Image intensity normalization was conducted using Z-score normalization to standardize voxel intensity distributions and reduce inter-patient intensity variability. All images with different sequences were resampled to an isotropic voxel size of $1 \times 1 \times 1 \text{ mm}^3$. This process ensured consistent voxel dimensions, matrix size, and spatial correspondence among the multiparametric MRI datasets before radiomic feature extraction.

Gray-level discretization was conducted using the absolute discretization method with a fixed number of gray levels (NbGrey = 64) and min-max intensity bounds. Three-dimensional texture calculation was performed using a neighborhood distance of 1 voxel (DistanceWithNeighbours = 1). The largest connected ROI component was retained (KeepTheLargestCluster = true), while ROI union was disabled (isEnabledRoiUnion = false).

The aligned lesion segmentation masks described in Section 2.3 were applied to the corresponding preprocessed MRI images to define the lesion regions for subsequent radiomic analysis.

2.5. Radiomic Feature Extraction

Radiomic features were extracted using LIFEx software V25.06.1 [21]. The preprocessed MRI datasets and corresponding aligned lesion segmentation masks were imported into LIFEx V25.06.1 for quantitative feature calculation. A total of 144 radiomic features were extracted from each MRI sequence, including first-order intensity features, shape features, and texture features. LIFEx enables calculation of histogram-based intensity descriptors, morphological characteristics, and texture features derived from gray-level matrices to characterize intralesional heterogeneity. The extracted features were based on standardized radiomic feature definitions consistent with the recommendations of the Image Biomarker Standardisation Initiative (IBSI), enabling reproducible calculation of quantitative imaging biomarkers across different studies [22].

The extracted feature categories included:

- First-order intensity features, describing the statistical distribution of voxel intensities within the lesion, including measures such as mean, median, minimum, maximum, standard deviation, skewness, kurtosis, and histogram-based descriptors.
- Shape features, describing the three-dimensional geometric characteristics of the segmented lesion, including parameters related to lesion volume, surface area, compactness, and sphericity.
- Texture features, quantifying spatial relationships and heterogeneity of voxel intensities using gray-level-based matrices, including features derived from the gray-level co-occurrence matrix (GLCM), gray-level run-length matrix (GLRLM), gray-level zone-length matrix (GLZLM/GLSZM), and neighborhood gray-level difference matrix (NGLDM).

Radiomic features were extracted independently from the five MRI datasets included in the analysis: ADC, DCE-Pre, DCE-Post1, DCE-Post2, and T2W. The generated sequence-specific feature datasets were subsequently used for feature selection and ML model development.

2.6. Radiomic Feature Selection

Radiomic feature selection was conducted to reduce feature dimensionality, minimize redundancy among highly correlated predictors, and improve model generalizability, particularly for multi-sequence datasets where the number of extracted features substantially exceeded the number of available lesions.

All feature selection procedures were incorporated within the training process of the patient-aware nested cross-validation framework to prevent information leakage. Feature selection was conducted exclusively using the

training folds, and the selected features were subsequently applied unchanged to the corresponding validation and test folds.

Three sequential feature-selection strategies were investigated:

1. Correlation filtering followed by minimum redundancy maximum relevance (mRMR)
2. Elastic Net regularization followed by mRMR
3. mRMR followed by Elastic Net regularization

Correlation filtering was applied to identify and reduce highly correlated predictors. Features with an absolute Pearson correlation coefficient greater than 0.90 were considered redundant. Within each correlated group, the retained feature was selected based on its univariate discriminatory ability measured by the direction-independent ROC-AUC.

The mRMR algorithm was used to select features with high relevance to the target classification task while minimizing redundancy among selected variables. Feature relevance was estimated using mutual information between individual features and the binary outcome label, while redundancy was quantified based on correlations among selected features.

Elastic Net feature selection was implemented using regularized logistic regression, combining L1 and L2 penalties to achieve sparse feature selection while maintaining stability in the presence of correlated radiomic predictors. Features were ranked according to the absolute magnitude of their learned coefficients.

Because the optimal number of radiomic features may vary according to the dataset and feature-fusion strategy, the number of retained features was automatically optimized within the inner cross-validation loop. Candidate feature subset sizes were evaluated using a lightweight logistic regression model based on inner-fold ROC-AUC performance, and the feature number providing the highest validation performance was selected. When multiple feature numbers achieved similar performance, the smaller feature subset was preferred to improve model simplicity and reduce overfitting risk.

2.7. ML Model Development

ML models were developed to classify breast lesions into benign and malignant categories using the selected radiomic features. The lesion ROI was considered the prediction unit; however, because some patients contributed more than one lesion, patient identity was used as the grouping variable throughout model development and validation to prevent patient-level data leakage.

All ML analyses were implemented in Python version 3.13.9. The complete computational workflow, including feature preprocessing, feature selection, model training, hyperparameter optimization, CV, and performance evaluation, was conducted using Python-based ML libraries.

Model development was conducted using a patient-aware nested CV framework. The outer loop consisted of five CV folds and was used for unbiased estimation of model performance on unseen patients, whereas a three-fold inner loop was used for feature-selection optimization, determination of the optimal feature-set size, class-

balancing evaluation, and hyperparameter optimization. All data-processing steps were restricted to the corresponding training folds to prevent information leakage.

Within each training partition, missing values were handled using median imputation derived from the training data, followed by variance filtering and feature standardization. The calculated transformation parameters were subsequently applied unchanged to the corresponding validation and test datasets. Feature selection and optimization procedures were conducted only within the training partitions.

A total of ten supervised ML classifiers representing different modelling approaches were evaluated, including Logistic Regression, linear Support Vector Machine (SVM), radial basis function (RBF)-SVM, Random Forest, Extra Trees, Gradient Boosting, XGBoost, LightGBM, K-Nearest Neighbors (KNN), and Gaussian Naive Bayes. These models were selected to provide representation from linear, kernel-based, tree-based ensemble, distance-based, and probabilistic learning approaches.

Hyperparameter optimization was conducted within the inner CV loop using Optuna with a tree-structured Parzen estimator (TPE) optimization strategy. Hyperparameters were optimized independently for each classifier, MRI feature configuration, and feature-selection strategy to identify the optimal model parameters without accessing outer test data. The predefined search spaces for each classifier, including continuous, categorical, and logarithmically sampled parameters, are summarized in Table 2. Logarithmic sampling was applied for parameters spanning several orders of magnitude to improve optimization efficiency.

Because the difference between benign and malignant samples was limited, class imbalance correction was not applied universally. Instead, the Synthetic Minority Oversampling Technique (SMOTE) was evaluated as an additional modelling condition. When applied, SMOTE was conducted only after feature selection and feature standardization and was restricted exclusively to training samples.

The complete modelling procedure was repeated across all evaluated MRI feature configurations. Radiomic predictors from matched lesion ROIs were combined to generate single-sequence and multi-sequence feature-fusion datasets, including different MRI source combinations (single-source, pairwise, triple-source, quadruple-source, and all-source configurations). The optimized models generated out-of-fold predictions from held-out outer-test patients, which were used for final performance assessment, as described in the next section.

An overview of the complete computational workflow, including MRI feature configuration generation, patient-aware nested CV, training pipeline, optimization procedures, and model evaluation, is presented in Figure 3.

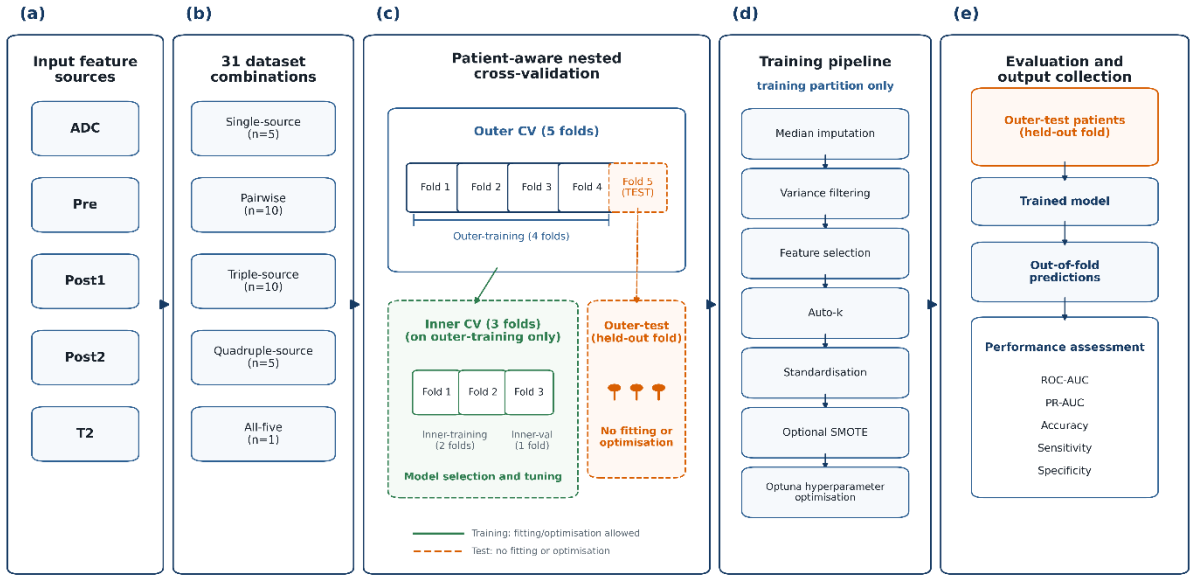


Figure 3. Overview of the patient-aware nested cross-validation workflow for machine-learning model development. MRI radiomic feature configurations were evaluated using nested cross-validation, including training-only preprocessing, feature selection, hyperparameter optimization, and evaluation on held-out patients.

Table 2. Hyperparameter search spaces used for Optuna-based optimization of machine-learning classifiers.

Classifier	Hyperparameter	Search space
Logistic Regression (Elastic-Net)	C	10^{-4} –1 (log)
	l1_ratio	0–1
Linear SVM	C	10^{-4} –1 (log)
RBF-SVM	C	10^{-3} –10 (log)
	γ	10^{-4} – 10^{-1} (log)
Random Forest	n_estimators	50, 100, 150, 200
	max_depth	2–8
	min_samples_leaf	1–10
	max_features	sqrt, log2, all features
Extra Trees	n_estimators	50–500 (step = 50)
	max_depth	2–10
	min_samples_leaf	2–20
Gradient Boosting	n_estimators	25, 50, 75, 100
	max_depth	1–3
	learning_rate	0.01–0.10 (log)
K-Nearest Neighbors (KNN)	n_neighbors	3–15
	weights	Uniform, distance
	metric	Euclidean, Manhattan
Gaussian Naive Bayes	var_smoothing	10^{-12} – 10^{-6} (log)
XGBoost	n_estimators	50–300 (step = 50)

LightGBM	max_depth	2–8
	learning_rate	0.01–0.30 (log)
	subsample	0.60–1.00
	colsample_bytree	0.50–1.00
	reg_alpha	10 ⁻⁸ –10 (log)
	reg_lambda	10 ⁻⁸ –10 (log)
	n_estimators	50–300 (step = 50)
	max_depth	2–8
	num_leaves	8–64
	learning_rate	0.01–0.30 (log)
	subsample	0.60–1.00
	colsample_bytree	0.50–1.00
	reg_alpha	10 ⁻⁸ –10 (log)
	reg_lambda	10 ⁻⁸ –10 (log)

*Log indicates logarithmic sampling during Optuna optimization. Continuous parameters were optimized within the specified ranges, whereas categorical parameters were selected from the listed candidate values.

2.8. ML Model Evaluation

Model evaluation was conducted using predictions generated exclusively from the outer test folds of the patient-aware nested CV framework. Each ROI received a single out-of-fold prediction generated by a model that had not been trained using that ROI or any other ROI from the same patient, ensuring unbiased estimation of generalization performance.

The primary evaluation metric was the mean ROC-AUC across the five outer folds, with the corresponding standard deviation reported to demonstrate performance variability among different patient subsets. Additional discrimination and classification metrics were calculated, including precision-recall AUC (PR-AUC), accuracy, balanced accuracy, sensitivity, specificity, precision, F1-score, and Cohen’s kappa coefficient.

Continuous model outputs were converted into binary predictions using a predefined probability threshold of 0.5. The resulting predictions were used to generate confusion matrices and calculate threshold-dependent classification metrics.

To provide an overall assessment of model performance, out-of-fold predictions from all outer folds were combined to generate pooled ROC curves, precision-recall curves, confusion matrices, and calibration analyses. Model calibration was evaluated by comparing predicted probabilities with observed class frequencies to assess the agreement between model confidence and actual outcomes. Calibration error was summarized using the Brier score. Lower Brier scores indicate better agreement between predicted probabilities and observed outcomes.

Uncertainty estimation was conducted using patient-level bootstrap resampling. Patients, rather than individual ROIs, were sampled with replacement to preserve lesion clustering within patients and avoid underestimation of uncertainty due to correlated observations. A total of $B = 2000$ bootstrap samples were generated from the available patient cohort.

For each bootstrap iteration, performance metrics were recalculated on the resampled dataset. Bootstrap samples containing only a single class were excluded when the corresponding metric could not be reliably computed. The 95% confidence intervals (CIs) were estimated using the percentile bootstrap method, where the lower and upper limits were defined as the 2.5th and 97.5th percentiles of the bootstrap distribution. CIs were calculated for pooled performance measures, including ROC-AUC, PR-AUC, accuracy, balanced accuracy, precision, sensitivity, specificity, F1-score, Cohen's kappa, and Brier score.

Candidate models were primarily ranked according to the mean ROC-AUC obtained from the outer cross-validation folds. However, final model comparison considered pooled bootstrap-corrected performance estimates, fold-to-fold variability, and calibration characteristics to provide a comprehensive assessment of model robustness and generalizability.

2.9. Clinical Interpretation and Clinical Impact Outcomes

To evaluate the interpretability and potential clinical relevance of the developed radiomics models, additional analyses were conducted on the selected models. Feature-selection stability was assessed across outer CV folds to determine the reproducibility of selected radiomic biomarkers. The selection frequency of each feature was calculated based on the number of folds in which it was retained, allowing identification of consistently selected radiomic characteristics.

Model interpretation was conducted using SHAP to quantify the contribution of individual radiomic features to malignant-class predictions. SHAP analysis was applied to recurrently selected features, and global feature importance was calculated using mean absolute SHAP values. SHAP summary plots were generated to demonstrate both the magnitude and direction of feature contributions across lesion ROIs.

The SHAP values were interpreted as measures of model behaviour and feature contribution rather than evidence of direct biological causality. Features consistently influencing model predictions were considered potential imaging biomarkers associated with lesion classification.

From a clinical perspective, the developed framework was designed as a quantitative decision-support approach to complement radiological interpretation of breast MRI findings. By integrating multiparametric MRI-derived radiomic information, the model aimed to provide additional objective information for differentiation between benign and suspicious lesions. Clinical applicability was considered based on model discrimination performance, robustness across patient groups, calibration reliability, and interpretability of predictions. Further prospective validation and assessment in independent clinical cohorts would be required before implementation into routine breast MRI workflows.

3. RESULTS

3.1. Dataset Characteristics and Quality-Control Assessment

A total of 31 radiomic feature configurations were evaluated, including five single-source datasets, ten pairwise combinations, ten triple-source combinations, five quadruple-source combinations, and one complete five-source dataset. The single-source datasets included radiomic features derived from ADC, DCE-Pre, DCE-Post1, DCE-

Post2, and T2W. These datasets represented 48 unique patients and included 117 lesion ROIs, with 144 numerical radiomic features extracted from each source. The distribution of benign and malignant ROIs was relatively balanced across the individual datasets, comprising 52 class 0 (benign) and 65 class 1 (malignant) ROIs. Patients contributed between one and seven lesions.

Potential outliers identified during quality-control analysis were retained because the purpose of this step was descriptive assessment rather than data removal. Missing values were subsequently handled within the training folds using median imputation to prevent information leakage. All ROIs from the same patient remained within the same CV partition throughout model development. The complete five-source dataset contained 117 matched ROIs and 720 radiomic predictors.

3.2. Overall Performance Across All Radiomic Feature Configurations

All 31 feature configurations were evaluated using the same patient-aware nested CV framework. For each configuration, the optimal pipeline was selected according to the highest mean outer-fold ROC-AUC across five patient-level validation folds.

Figure 4 summarizes the classification performance of the best-performing pipeline for each radiomic configuration. Error bars represent the standard deviation (SD) of ROC-AUC values across the five outer folds. The figure presents results separately for single-source (Figure 4a), pairwise (Figure 4b), triple-source (Figure 4c), quadruple-source (Figure 4d), and the best-performing configuration at each fusion level (Figure 4e).

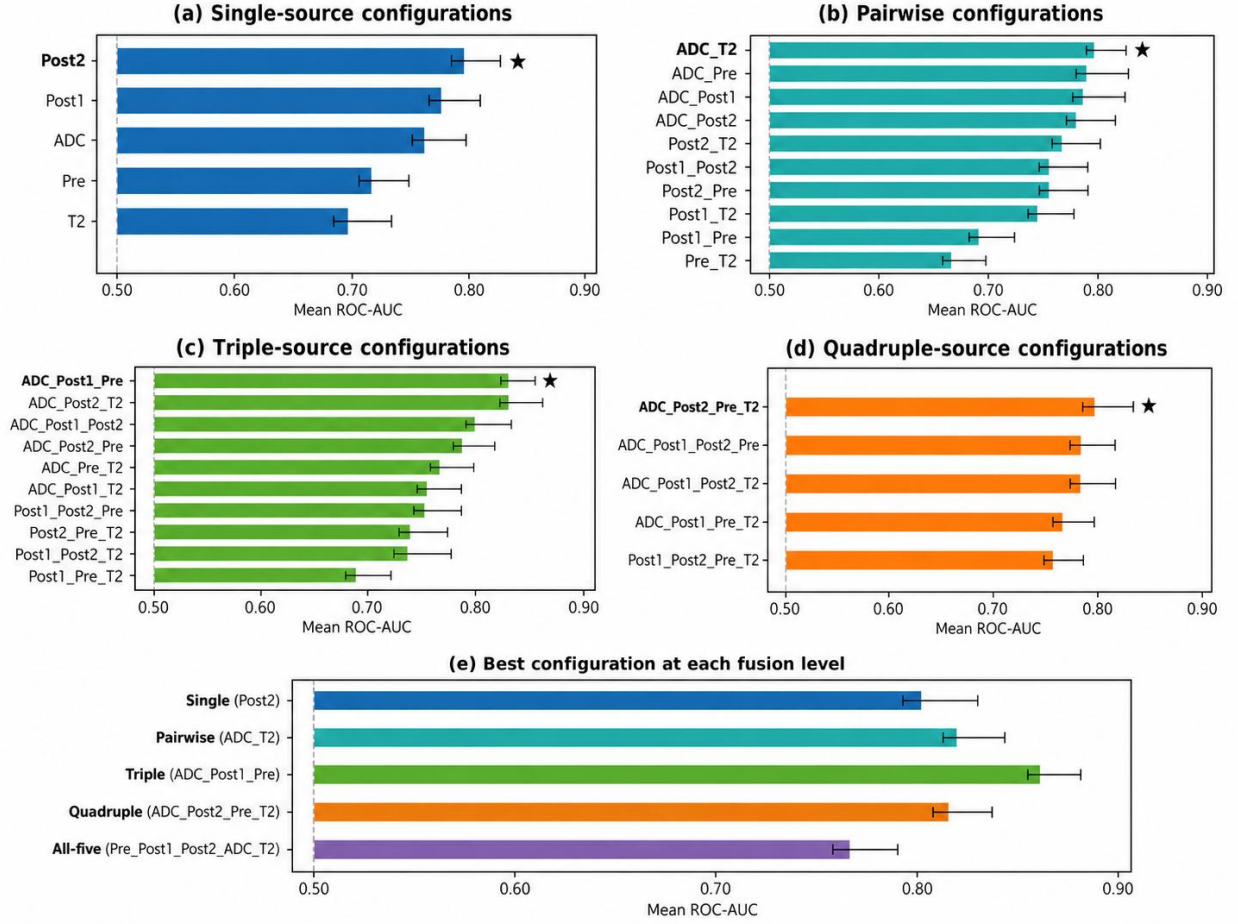


Figure 4. Classification performance across the 31 radiomic source configurations. The best pipeline for each configuration is shown according to mean ROC- AUC across five patient- aware outer folds. Error bars represent the corresponding standard deviation. **(a)** Single- source, **(b)** pairwise, **(c)** triple- source, **(d)** quadruple- source configurations, and **(e)** the best configuration at each fusion level. The dashed line indicates a chance-level ROC- AUC of 0.50.

The highest overall classification performance was obtained using the ADC+Post1 DCE+Pre DCE triple-source configuration. This model used Elastic Net followed by mRMR feature selection, no SMOTE balancing, and a linear support vector machine (SVM) classifier as the highest-performing model. It achieved a mean outer-fold ROC-AUC of 0.836 ± 0.088 with a pooled ROC-AUC of 0.799. Auto-k feature selection retained an **average of 10 features** across the five outer folds.

The ADC–Post2–T2 triple-source configuration showed comparable performance with a mean ROC-AUC of 0.834 ± 0.045 , demonstrating lower fold-to-fold variability, although its pooled ROC-AUC was lower (0.756).

Among different fusion levels, the highest-ranked single-source configuration was Post2, the highest-ranked pairwise configuration was ADC+T2, the highest-ranked quadruple-source configuration was ADC+DCE Post2+DCE Pre+T2, and the complete five-source configuration did not achieve the highest performance.

These findings indicate that increasing the number of MRI sources did not result in continuous improvement of classification performance. The highest-performing model was obtained from a selected three-source combination

rather than the complete multiparametric feature set. The best-performing pipeline at each fusion level is summarized in Table 3.

Table 3. Highest- ranked configuration at each feature- fusion level. Configurations were selected according to mean ROC- AUC across the five patient- aware outer folds. Pooled ROC- AUC was calculated from the combined out-of-fold predictions.

Fusion level	Source configuration	Feature selection	SMOTE	Classifier	Mean AUC $\pm$ SD	Pooled Mean AUC	Mean k*
Single	Post2	EN $\rightarrow$ mRMR	No	RBF SVM	0.790 $\pm$ 0.127	0.744	10.8 ($\approx$ 11)
Pairwise	ADC+T2W	EN $\rightarrow$ mRMR	No	RBF SVM	0.799 $\pm$ 0.084	0.722	10.4 ($\approx$ 10)
Triple	ADC+Post1+Pre	EN $\rightarrow$ mRMR	No	Linear SVM	0.836 $\pm$ 0.088	0.799	9.6 ($\approx$ 10)
Quadruple	ADC+Post2-Pre+T2W	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.802 $\pm$ 0.096	0.723	13.2 ($\approx$ 13)
Five-source	ADC+Pre-Post1+Post2+T2W	EN $\rightarrow$ mRMR	No	RBF SVM	0.747 $\pm$ 0.117	0.724	12.4 ($\approx$ 12)

Abbreviations: ADC, apparent diffusion coefficient; AUC, area under the receiver operating characteristic curve; EN, elastic net; k, number of selected radiomic features; mRMR, minimum redundancy maximum relevance; Post1, first post-contrast sequence; Post2, second post-contrast sequence; Pre, pre-contrast sequence; RBF, radial basis function; SD, standard deviation; SMOTE, synthetic minority oversampling technique; SVM, support vector machine; T2W, T2-weighted sequence.

*Mean k values were rounded to the nearest integer.

Figure 5 illustrates the average impact of SMOTE on model performance across different feature-fusion levels. The changes in mean outer-fold ROC-AUC and pooled balanced accuracy after applying SMOTE are shown for single-, pairwise-, triple-, quadruple-, and five-source fusion configurations. Overall, SMOTE had a limited effect on discriminative performance, with only small changes observed in mean outer-fold ROC-AUC (ranging from -0.007 to $+0.003$). In contrast, pooled balanced accuracy consistently improved after SMOTE application across all fusion levels, with increases ranging from $+0.010$ to $+0.019$. The greatest improvement in pooled balanced accuracy was observed for pairwise fusion ($+0.019$), followed by single-source fusion ($+0.015$) and five-source fusion ($+0.014$). However, the five-source fusion configuration showed a decrease in mean outer-fold ROC-AUC (-0.007), suggesting that SMOTE improved class balance and overall classification accuracy but did not necessarily enhance model discrimination in more complex feature-fusion settings.

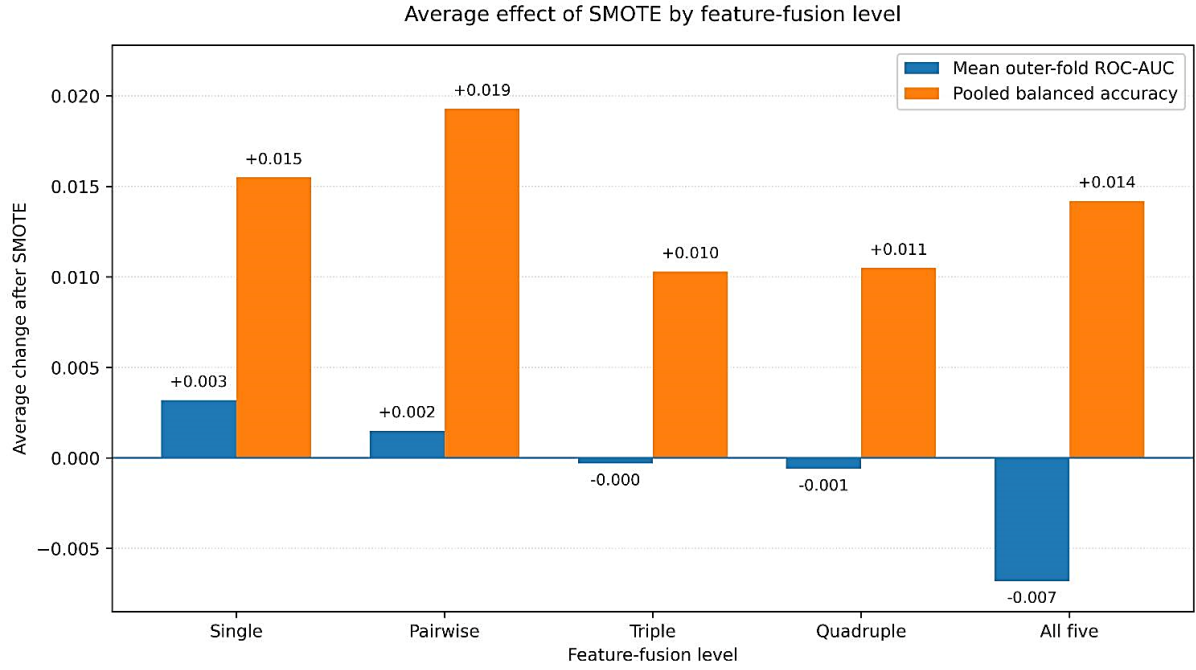


Figure 5. Best-Performing Pipeline at Each Feature-Fusion Level.

3.3. Single-Source Radiomic Classification Performance

The five single-source datasets achieved mean outer-fold ROC-AUC values ranging from 0.674 to 0.790 (Figure 3a). Among these, the Post2 dataset yielded the best-performing single-source configuration. The highest-ranked Post2 pipeline consisted of Elastic Net feature selection followed by mRMR, no SMOTE application, and an RBF SVM classifier. This configuration achieved a mean outer-fold ROC-AUC of 0.790 ± 0.127 and a pooled ROC-AUC of 0.744. The corresponding mean and pooled PR-AUC values were 0.876 and 0.815, respectively.

The Post1 dataset demonstrated the second-highest-performing single-source performance, with a mean outer-fold ROC-AUC of 0.767 ± 0.098 , while achieving the highest pooled ROC-AUC among all single-source datasets (0.765). The ADC, Pre, and T2 datasets achieved mean ROC-AUC values of 0.764, 0.698, and 0.674, respectively. Among the five highest-ranked single-source pipelines, four incorporated SMOTE, whereas the Post2 optimal pipeline was the only configuration that achieved superior performance without oversampling. Detailed results for all evaluated pipelines are provided in Table 4.

Table 4. Highest- ranked pipeline for each single- source dataset. Pipelines were ranked by mean ROC- AUC across five patient-aware outer folds.

Dataset	Feature Selection	SMOTE	Classifier	Mean AUC $\pm$ SD	Pooled AUC	Mean PR-AUC	Pooled PR-AUC	Pooled Bal. Acc.
Post2	EN $\rightarrow$ mRMR	No	RBF SVM	0.790 ± 0.127	0.744	0.876	0.815	0.669
Post1	EN $\rightarrow$ mRMR	Yes	KNN	0.767 ± 0.098	0.765	0.798	0.815	0.705

ADC	mRMR → EN	Yes	Linear SVM	0.764 ± 0.080	0.691	0.799	0.720	0.683
Pre	mRMR → EN	Yes	Logistic Regression	0.698 ± 0.070	0.679	0.753	0.740	0.594
T2W	EN → mRMR	Yes	RBF SVM	0.674 ± 0.049	0.642	0.708	0.679	0.590

Abbreviations: ADC, apparent diffusion coefficient; AUC, area under the receiver operating characteristic curve; Bal. Acc., balanced accuracy; EN, elastic net; KNN, k-nearest neighbors; mRMR, minimum redundancy maximum relevance; PR-AUC, area under the precision–recall curve; RBF, radial basis function; SD, standard deviation; SMOTE, synthetic minority oversampling technique; SVM, support vector machine; k, number of selected features; Post1, first post-contrast phase; Post2, second post-contrast phase; Pre, pre-contrast phase; T2W, T2-weighted sequence.

3.4. Pairwise Feature-Fusion Performance

Ten pairwise MRI source combinations were evaluated to investigate whether combining complementary radiomic information improved lesion classification. The mean outer-fold ROC-AUC values ranged from 0.665 to 0.799 across the pairwise configurations (Figure 3b).

The highest-performing pairwise configuration was ADC–T2, which achieved a mean ROC-AUC of 0.799 ± 0.084 using Elastic Net followed by mRMR feature selection, without SMOTE, and an RBF-SVM classifier. The pooled ROC-AUC for this model was 0.722.

The ADC–Pre combination ranked second with a mean ROC-AUC of 0.792 ± 0.035 , showing the lowest variability among the pairwise winning models. ADC–Post1 ranked third with a mean ROC-AUC of 0.788 ± 0.050 and achieved the highest pooled ROC-AUC among pairwise configurations (0.775).

Notably, all four highest-ranked pairwise configurations included ADC features, suggesting that diffusion-derived quantitative information provided complementary predictive value when combined with other MRI sources. Elastic Net followed by mRMR was selected in seven of the ten pairwise configurations, and seven optimal pairwise pipelines incorporated SMOTE. The complete pairwise results are summarized in Table 5.

Table 5. Highest- ranked pipeline for each pairwise source configuration. Pipelines were ranked by mean ROC-AUC across five patient- aware outer folds.

Configuration	Feature selection	SMOTE	Classifier	Mean AUC ± SD	Pooled AUC	Mean k
ADC+T2W	EN → mRMR	No	RBF SVM	0.799 ± 0.084	0.722	10.4 (≈10)
ADPre	EN → mRMR	No	Logistic Regression	0.792 ± 0.035	0.722	9.6 (≈10)
AD+-Post1	EN → mRMR	Yes	RBF SVM	0.788 ± 0.050	0.775	8.0
ADC+Post2	mRMR → EN	Yes	Linear SVM	0.777 ± 0.101	0.687	8.4 (≈8)
Post2+T2W	mRMR → EN	Yes	XGBoost	0.771 ± 0.074	0.743	8.4 (≈8)
Post1+Post2	mRMR → EN	Yes	RBF SVM	0.763 ± 0.124	0.744	5.6 (≈5)
Post2+Pre	EN → mRMR	Yes	RBF SVM	0.756 ± 0.147	0.748	10.4 (≈10)

Post1+T2W	EN → mRMR	Yes	KNN	0.742 ± 0.112	0.718	11.2 (≈11)
Post1+Pre	EN → mRMR	Yes	Logistic Regression	0.694 ± 0.138	0.687	10.4 (≈10)
Pre+T2W	EN → mRMR	No	Gaussian NB	0.665 ± 0.048	0.633	9.6 (≈10)

Abbreviations: ADC, apparent diffusion coefficient; AUC, area under the receiver operating characteristic curve; Bal; EN, elastic net; mRMR, minimum redundancy maximum relevance; RBF, radial basis function; SD, standard deviation; SMOTE, synthetic minority oversampling technique; SVM, support vector machine; k, number of selected features; Post1, first post-contrast phase; Post2, second post-contrast phase; Pre, pre-contrast phase; T2W, T2-weighted sequence.

3.5. Triple-Source Feature-Fusion Performance

The ten triple-source configurations demonstrated mean outer-fold ROC-AUC values ranging from 0.682 to 0.836 (Figure 3c). The best-performing model across all evaluated feature configurations was obtained from the ADC+Post1 DCE+Pre DCE triple-source combination. The optimal pipeline consisted of Elastic Net followed by mRMR feature selection, no SMOTE, and a linear SVM classifier. This model achieved a mean outer-fold ROC-AUC of 0.836 ± 0.088 and a pooled ROC-AUC of 0.799.

The corresponding mean and pooled PR-AUC values were 0.879 and 0.832, respectively, while the pooled balanced accuracy was 0.713. The ADC+Post2 DCE+T2 configuration achieved a comparable mean ROC-AUC of 0.834 ± 0.045 , with improved consistency across outer folds, although its pooled ROC-AUC was lower (0.756). ADC+Post1 DCE+Post2 DCE ranked third among triple-source configurations, achieving a mean ROC-AUC of 0.797 ± 0.075 and pooled ROC-AUC of 0.790.

Across triple-source configurations, ADC was present in all six highest-ranked combinations. Elastic Net followed by mRMR remained the most frequently selected feature-selection strategy, appearing in six of ten optimal pipelines. SMOTE was selected in half of the triple-source winning models. The detailed triple-source comparison is provided in Table 6.

Table 6. Highest-ranked pipeline for each triple-source configuration. Pipelines were ranked by mean ROC-AUC across five patient-aware outer folds.

Configuration	Feature selection	SMOTE	Classifier	Mean AUC ± SD	Pooled AUC	Mean k
ADC+Post1+Pre	EN → mRMR	No	Linear SVM	0.836 ± 0.088	0.799	9.6 (≈10)
ADC+Post2+T2W	EN → mRMR	No	Linear SVM	0.834 ± 0.045	0.756	10.4 (≈10)
ADC+Post1+Post2	EN → mRMR	Yes	Logistic Regression	0.797 ± 0.075	0.790	7.6 (≈8)
ADC+Post2+Pre	mRMR → EN	Yes	Logistic Regression	0.793 ± 0.050	0.722	14.4 (≈14)

ADC+Pre+T2W	EN → mRMR	Yes	LightGBM	0.767 ± 0.115	0.720	13.2 (≈13)
ADC+Post1+T2W	EN → mRMR	Yes	Logistic Regression	0.754 ± 0.158	0.765	5.6 (≈5)
Post1+Post2+Pre	mRMR → EN	Yes	Linear SVM	0.753 ± 0.126	0.715	11.6 (≈12)
Post2+Pre+T2W	EN → mRMR	No	RBF SVM	0.735 ± 0.176	0.707	13.2 (≈13)
Post1+Post2+T2W	mRMR → EN	No	Gaussian NB	0.735 ± 0.057	0.706	10.0
Post1+Pre+T2W	mRMR → EN	No	Gaussian NB	0.682 ± 0.074	0.695	11.6 (≈12)

Abbreviations: ADC, apparent diffusion coefficient; AUC, area under the receiver operating characteristic curve; EN, elastic net; Gaussian NB, Gaussian naïve Bayes; KNN, k-nearest neighbors; LightGBM, light gradient boosting machine; mRMR, minimum redundancy maximum relevance; PR-AUC, area under the precision–recall curve; RBF, radial basis function; SD, standard deviation; SMOTE, synthetic minority oversampling technique; SVM, support vector machine; k, number of selected radiomic features; Post1, first post-contrast phase; Post2, second post-contrast phase; Pre, pre-contrast phase; T2W, T2-weighted sequence.

3.6. Quadruple-Source Feature-Fusion Performance

The five quadruple-source configurations produced mean outer-fold ROC-AUC values ranging from 0.756 to 0.802 (Figure 3d). The highest-performing quadruple-source model was obtained using ADC–Pre–Post2–T2 features. This pipeline used Elastic Net followed by mRMR feature selection, SMOTE balancing, and an RBF-SVM classifier. The model achieved a mean ROC-AUC of 0.802 ± 0.096 with a pooled ROC-AUC of 0.723.

The ADC–Pre–Post1–Post2 configuration ranked second with a mean ROC-AUC of 0.790 ± 0.063 , whereas ADC–Post1–Post2–T2 achieved the third-highest performance with a mean ROC-AUC of 0.788 ± 0.072 and the highest pooled ROC-AUC among quadruple-source models (0.770). All five optimal quadruple-source pipelines used Elastic Net followed by mRMR feature selection, and four incorporated SMOTE. The complete quadruple-source comparison is summarized in Table 7.

Table 7. Highest- ranked pipeline for each quadruple- source configuration. Pipelines were ranked by mean ROC-AUC across five patient- aware outer folds.

Configuration	Feature selection	SMOTE	Classifier	Mean AUC ± SD	Pooled AUC
ADC+Pre+Post2+T2	EN → mRMR	Yes	RBF SVM	0.802 ± 0.096	0.723
ADC+Pre+Post1+Post2	EN → mRMR	Yes	RBF SVM	0.790 ± 0.063	0.765
ADC+Post1+Post2+T2W	EN → mRMR	Yes	Logistic Regression	0.788 ± 0.072	0.770
ADC+Pre+Post1+T2W	EN → mRMR	No	Linear SVM	0.767 ± 0.063	0.725
Pre+Post1+Post2+T2W	EN → mRMR	Yes	RBF SVM	0.756 ± 0.184	0.713

Abbreviations: AUC, area under the receiver operating characteristic curve; EN, elastic net; mRMR, minimum redundancy maximum relevance; RBF, radial basis function; SD, standard deviation; SVM, support vector machine; T2W, T2-weighted imaging; Post1, first post-contrast phase; Post2, second post-contrast phase; Pre, pre-contrast phase.

3.7. Complete Five-Source Model Performance

The complete multiparametric MRI model incorporated ADC, DCE-Pre, DCE-Post1, DCE-Post2, and T2 radiomic features (Figure 3e). This dataset contained 114 matched ROIs and 720 predictor features.

The highest-ranked five-source pipeline consisted of Elastic Net followed by mRMR feature selection, no SMOTE, and RBF-SVM classification. The model achieved a mean outer-fold ROC-AUC of 0.747 ± 0.117 and a pooled ROC-AUC of 0.724. The mean and pooled PR-AUC values were 0.810 and 0.792, respectively. The pooled balanced accuracy was 0.595, with pooled sensitivity of 0.810, precision of 0.622, and F1-score of 0.703 at the predefined classification threshold of 0.5. The pooled Cohen's kappa and Brier score were 0.197 and 0.220, respectively. Auto-k feature selection retained an average of ≈ 12 features, while 38 different features were selected in at least one outer fold.

Although the five-source model incorporated the largest amount of MRI information, its performance remained lower than the highest-performing triple-source model. The mean ROC-AUC was 0.089 lower than the ADC–Post1–Pre configuration, indicating that inclusion of additional feature sources did not necessarily improve classification performance.

3.8. Comparison of Feature-Selection Strategies

Across the evaluated radiomic configurations, Elastic Net regularization followed by mRMR was the most frequently selected feature-selection strategy among the optimal pipelines. As shown in Figure 6a, the distribution of the best mean outer-fold ROC-AUC values varied according to the selected feature-selection strategy. The EN $\rightarrow$ mRMR approach demonstrated the highest overall performance distribution, with a higher median ROC-AUC and the highest average performance compared with the other strategies. The mRMR $\rightarrow$ EN strategy showed comparable but slightly lower performance, whereas Corr. $\rightarrow$ mRMR resulted in the lowest overall ROC-AUC distribution among the evaluated approaches. The mean performance of each strategy is indicated by the diamond markers.

Figure 6b summarizes the number of feature configurations for which each feature-selection strategy achieved the highest ranking across different MRI fusion levels. EN $\rightarrow$ mRMR was selected as the best-performing strategy in 20 of the 31 evaluated configurations, including single-source, pairwise, triple-source, quadruple-source, and complete five-source datasets. mRMR $\rightarrow$ EN ranked first in 11 configurations, whereas Corr. $\rightarrow$ mRMR did not achieve the highest rank in any evaluated configuration.

The frequent selection of Elastic Net–mRMR pipelines indicates that combining embedded regularization with relevance–redundancy-based feature ranking provided an effective strategy for reducing the high dimensionality of radiomic datasets while retaining informative predictors.

The number of retained radiomic features varied according to MRI feature configuration. In the best-performing ADC–Post1–Pre triple-source model, automatic feature-number optimization retained ≈ 10 features across the outer folds.

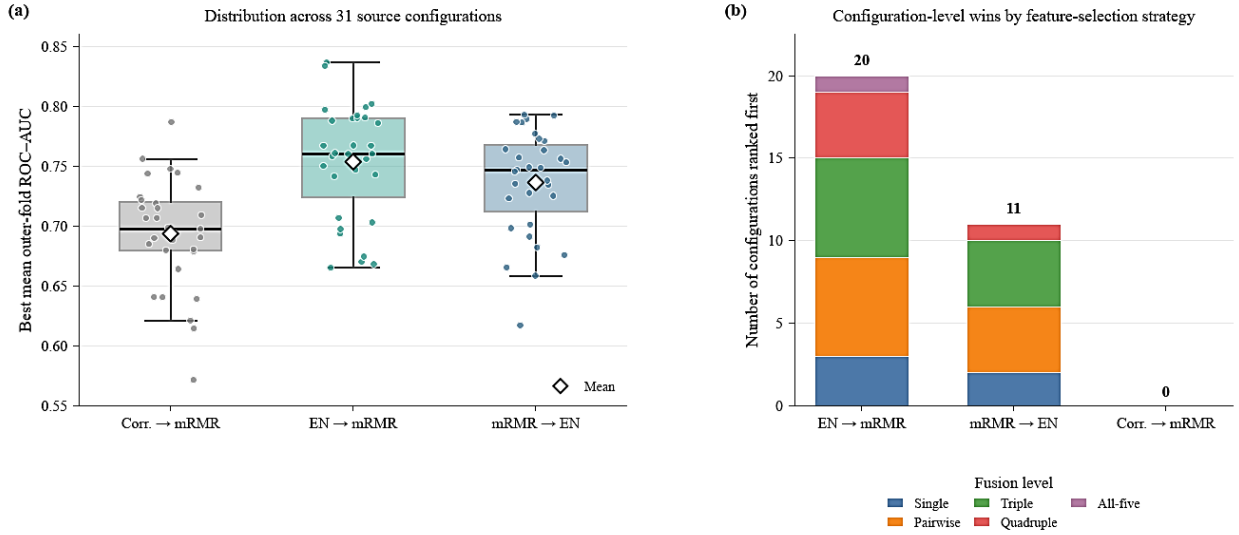


Figure 6. Comparison of feature-selection strategies across the 31 source configurations. **(a)** Distribution of the best mean outer-fold ROC-AUC obtained for each strategy. Points represent source configurations, boxes show the interquartile range, horizontal lines indicate medians, and white diamonds indicate means. **(b)** Number of configurations in which each strategy achieved the highest mean ROC-AUC, separated by feature-fusion level. EN, Elastic Net; mRMR, minimum redundancy maximum relevance.

3.9. Effect of SMOTE on Model Performance

Figure 7 illustrates the impact of SMOTE on mean outer-fold ROC-AUC across different feature-fusion configurations. Figure 7a shows the configuration-level difference in ROC-AUC between SMOTE and no-SMOTE models ($\Delta\text{ROC-AUC} = \text{SMOTE} - \text{no-SMOTE}$). The effect of SMOTE was heterogeneous, with changes ranging from -0.013 to $+0.025$. The largest improvements were observed for the triple-source Post1 + Post2 + Pre configuration ($+0.025$), followed by the single-source Post1 dataset ($+0.024$). Figure 7b summarizes the number of configurations favoring SMOTE at each fusion level. SMOTE showed the greatest advantage in single-source models (4/5 configurations), where benefit decreased with increasing fusion complexity, with the all-five-source configuration favoring no SMOTE.

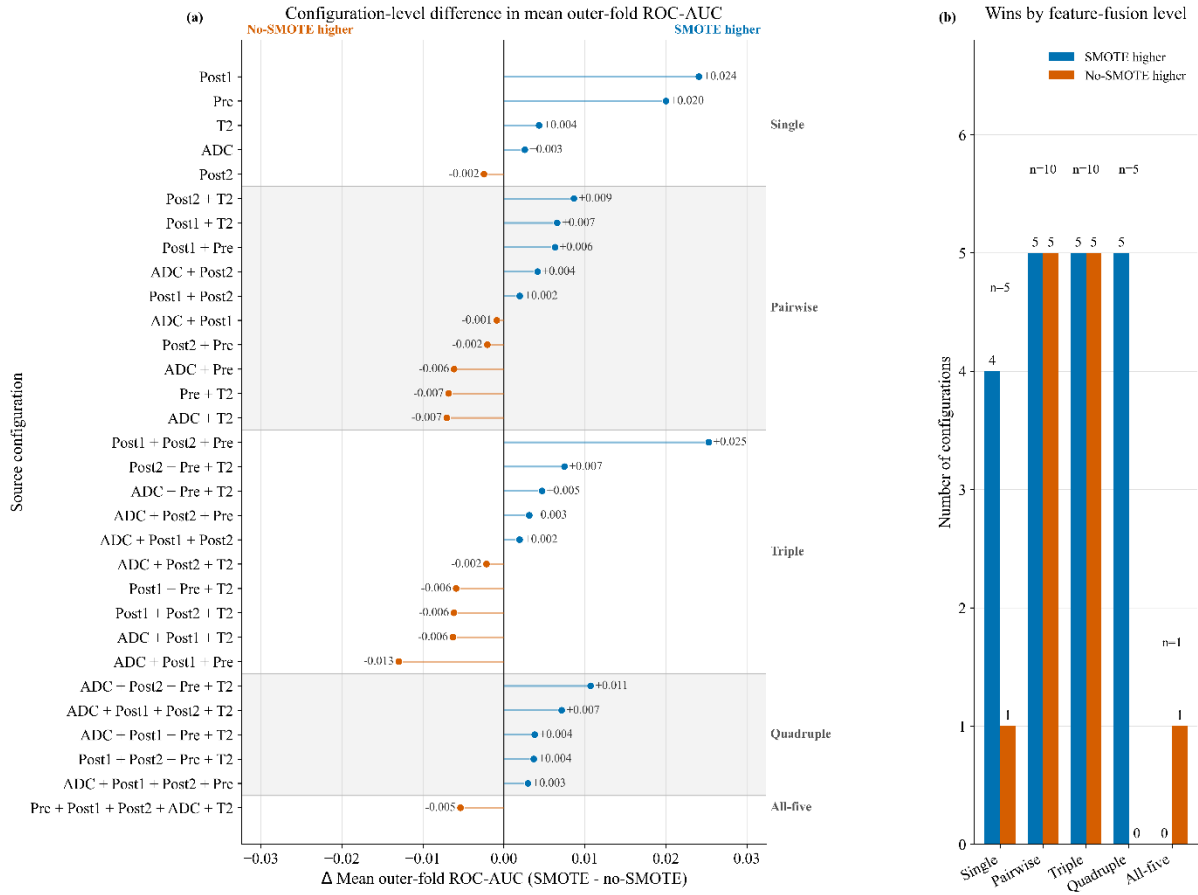


Figure 7. Impact of SMOTE on mean outer-fold ROC-AUC across feature-fusion configurations. **(a)** Configuration-level differences in mean outer-fold ROC-AUC between SMOTE and no-SMOTE models ($\Delta\text{ROC-AUC} = \text{SMOTE} - \text{no-SMOTE}$), with positive values indicating improved performance after SMOTE. **(b)** Number of configurations favoring SMOTE or no-SMOTE at each feature-fusion level. ADC, apparent diffusion coefficient; Post1, first post-contrast phase; Post2, second post-contrast phase; Pre, pre-contrast phase; T2, T2-weighted imaging; ROC-AUC, area under the receiver operating characteristic curve; SMOTE, synthetic minority oversampling technique.

3.10. Model Evaluation of the Best-Performing Pipeline

The frequency with which each classifier achieved the highest-ranked performance across the 31 radiomic source configurations is summarized in Figure 8. Among all evaluated classifiers, RBF-SVM was the most frequently selected optimal classifier, achieving the highest rank in 11 configurations, including single-source ($n=2$), pairwise ($n=4$), triple-source ($n=1$), quadruple-source ($n=3$), and five-source ($n=1$) feature configurations.

Logistic Regression was the second most frequently selected classifier, ranking first in seven configurations, followed by Linear SVM with six configurations. These linear and kernel-based models accounted for the majority of optimal pipelines across different MRI feature-fusion levels.

Probabilistic and distance-based classifiers showed fewer top-ranked configurations, with Gaussian Naive Bayes selected in 3 configurations and KNN in two configurations. Tree-based boosting models achieved the highest

rank less frequently, with XGBoost and LightGBM each selected in one configuration, while Random Forest, Extra Trees, and Gradient Boosting did not achieve the top rank in any evaluated configuration.

Overall, kernel-based SVM models, particularly RBF-SVM, demonstrated the most consistent performance across different radiomic feature configurations, whereas tree-based ensemble methods were less frequently selected as the optimal classifiers in this dataset.

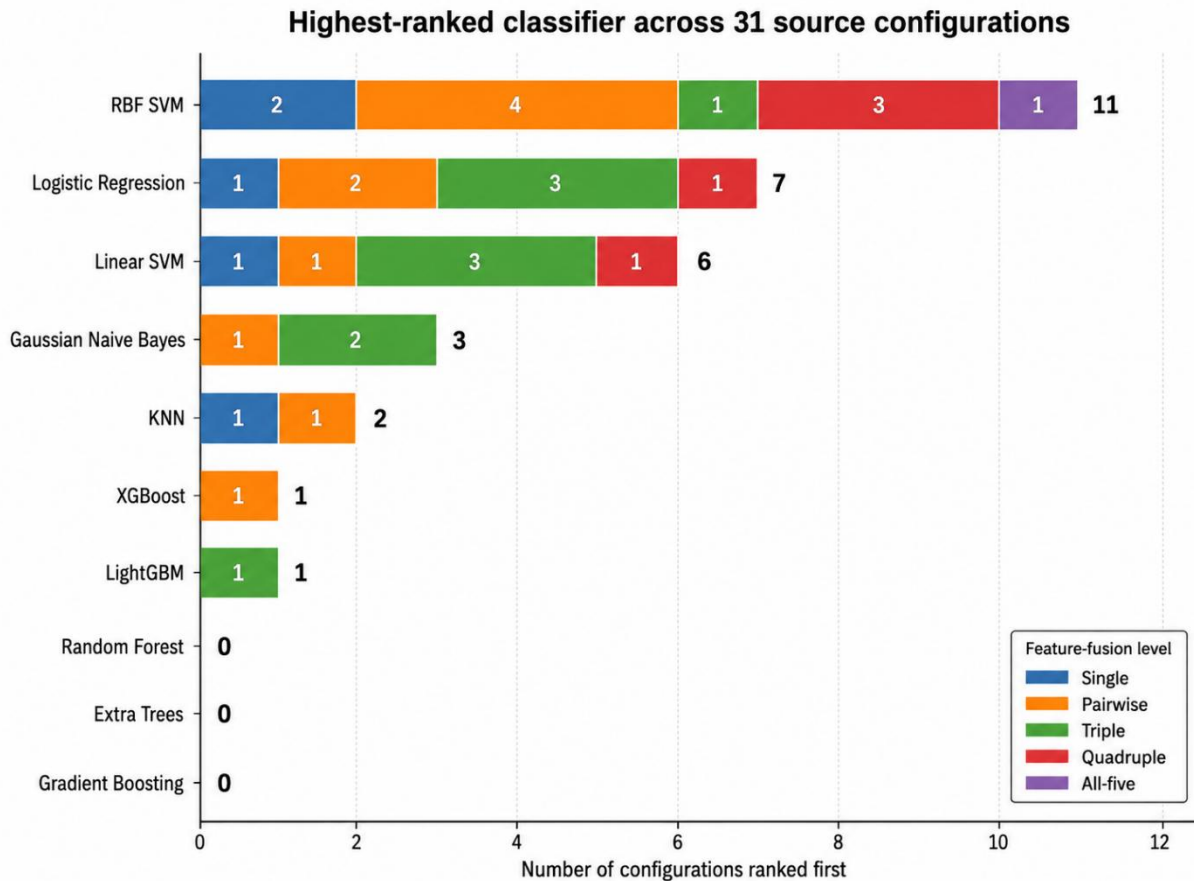


Figure 8. Distribution of the highest-ranked machine-learning classifiers across 31 MRI radiomic source configurations. The stacked bars represent the number of configurations in which each classifier achieved the best performance, stratified by feature-fusion level (single-source, pairwise, triple-source, quadruple-source, and complete five-source configurations).

The optimal model was obtained using the ADC–Post1–Pre radiomic configuration with Elastic Net → mRMR feature selection, Auto-k optimization, no SMOTE, and a linear SVM classifier. The pooled out-of-fold performance of this model is presented in Figure 9.

The model achieved a pooled ROC-AUC of 0.799 (Figure 9a) and an average precision of 0.832 on the precision–recall curve (Figure 9b). At a probability threshold of 0.5, the confusion matrix showed 47 true positives, 34 true negatives, 16 false positives, and 16 false negatives (Figure 9c). The calibration curve demonstrated agreement between predicted probabilities and observed outcomes, with a Brier score of 0.183 (Figure 9d).

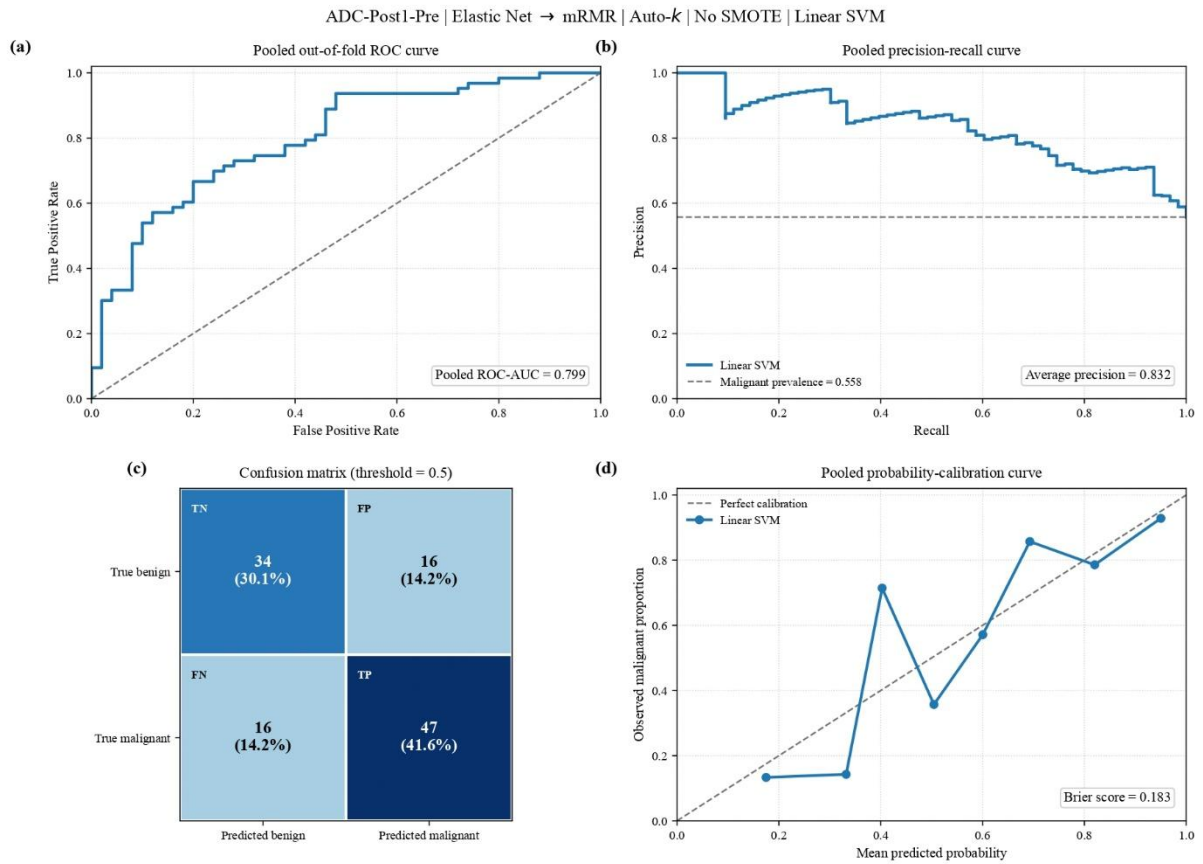


Figure 9. Detailed out-of-fold performance of the highest-ranked pipeline. The model used ADC, Post1, and Pre features with Elastic Net followed by mRMR, Auto-*k*, no SMOTE, and a linear SVM. **(a)** Pooled ROC curve. **(b)** Pooled precision-recall curve. **(c)** Confusion matrix at the predefined threshold of 0.5. **(d)** Probability-calibration curve. All results were calculated from pooled predictions across the five patient-aware outer-test folds. AUC, area under the curve; mRMR, minimum redundancy maximum relevance; SVM, support vector machine.

Calibration and decision curve analyses (DCA) were conducted to assess the reliability and potential clinical utility of the developed models beyond discrimination performance (Figure 10). The calibration plots (Figure 10a) demonstrated variability among classifiers in agreement between predicted probabilities and observed outcomes. The linear SVM model selected as the optimal classifier showed good probability calibration, with a relatively low Brier score compared with several alternative models, indicating better agreement between predicted malignancy risk and actual lesion classification.

DCA (Figure 10b) further demonstrated differences in potential clinical usefulness among models. The linear SVM model achieved the highest net benefit across a broad range of clinically relevant threshold probabilities compared with most alternative classifiers, suggesting that its predictions may provide greater potential utility for supporting individualized lesion assessment. These findings indicate that model evaluation based solely on ROC-AUC may not fully capture clinical applicability, and calibration and decision-analytic measures provide complementary information regarding model reliability and potential implementation.

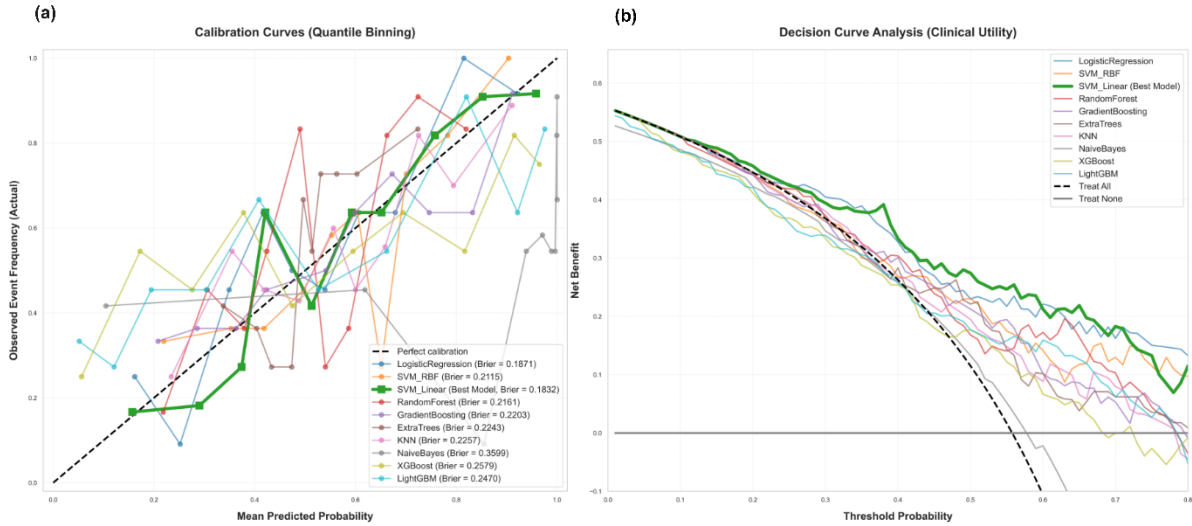


Figure 10. Calibration and clinical utility analysis of evaluated machine-learning classifiers. **(a)** Quantile-based calibration curves comparing predicted probabilities with observed event frequencies for all evaluated classifiers. The dashed diagonal line represents perfect calibration. Brier scores are provided for each model, with lower values indicating better agreement between predicted and observed probabilities. **(b)** Decision curve analysis demonstrating the net clinical benefit of each classifier across different threshold probabilities. The dashed and solid reference lines represent the strategies of treating all patients and treating none, respectively. The best-performing linear SVM model demonstrated favorable calibration and maintained higher clinical net benefit across a wider range of threshold probabilities compared with other evaluated classifiers.

3.11. Feature Stability and Model Interpretation

Feature stability was evaluated by recording the frequency with which individual radiomic features were selected across the outer cross-validation folds. In the optimal ADC-Post1-Pre model, the automatic feature-selection process retained an average of 9.6 features per fold. Across all outer folds, 28 unique radiomic features were selected at least once, while 13 features showed higher stability by being repeatedly selected in at least two outer folds.

Figure 11 provides an explainability analysis of the best-performing model and demonstrates the contribution, importance, and stability of the selected radiomic features. The SHAP summary plot (Figure 11a) indicates that the model prediction was mainly driven by ADC-derived first-order intensity features and Post1-DCE intensity and texture characteristics. Among all predictors, **ADC intensity histogram median features** showed the strongest contribution, with higher feature values generally associated with increased probability of malignancy. This finding is consistent with the biological relationship between reduced water diffusivity and increased cellular density in malignant lesions.

Post1-DCE features also demonstrated substantial influence on model prediction. First-order intensity descriptors, including histogram-based gradient and mode intensity features, contributed strongly to classification, suggesting that enhancement intensity distribution provided important information regarding lesion vascular characteristics and heterogeneity. In addition, morphological features derived from Post1-DCE images, including compactness

and sphericity, contributed to model discrimination, indicating that differences in lesion architecture and boundary characteristics were informative for distinguishing benign from malignant lesions.

The global SHAP importance ranking (Figure 11b) confirmed that ADC and Post1-DCE features represented the dominant predictors of the model. The highest-ranked features were primarily first-order intensity descriptors from ADC and DCE images, followed by morphological and texture-based descriptors from Post1. This demonstrates that the model relied on complementary information from diffusion characteristics, contrast enhancement behavior, and lesion structure rather than a single feature category.

Feature-selection recurrence analysis (Figure 11c) demonstrated the stability of the identified predictors across outer cross-validation folds. ADC histogram median intensity showed the highest recurrence frequency, followed by additional ADC and Post1-DCE intensity features. Several Post1 morphological and texture features were repeatedly selected across validation folds, suggesting that these features provided consistent predictive information rather than being selected due to random variation.

Overall, Figure 11 indicates that the machine-learning model used biologically interpretable MRI characteristics related to tissue diffusivity, enhancement behavior, and lesion morphology. The combination of SHAP contribution analysis and feature-selection stability assessment supports the transparency and reproducibility of the developed radiomic model.

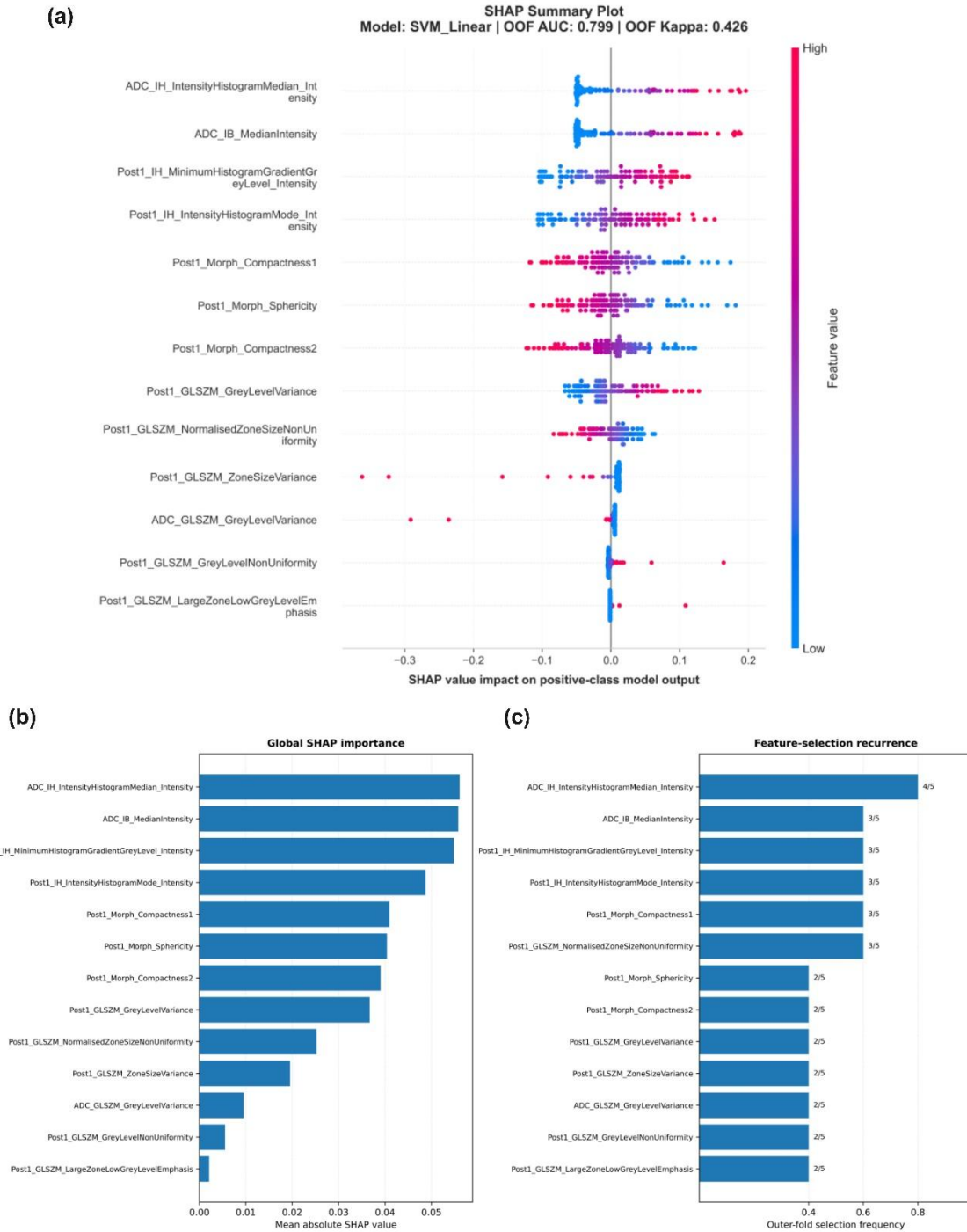


Figure 11. Explainability analysis of the optimal radiomic model using SHapley Additive exPlanations (SHAP). (a) SHAP summary plot demonstrating the contribution and directionality of the selected radiomic features on the model output. Each point represents one lesion, with red and blue colors indicating high and low feature values, respectively. Positive SHAP values indicate increased probability of malignancy, whereas negative values indicate contribution toward benign classification. (b) Global SHAP importance ranking based on mean absolute SHAP values, showing the overall contribution of each feature to model prediction. (c) Feature-selection recurrence across outer cross-validation folds, demonstrating the stability and reproducibility of selected radiomic predictors.

4. DISCUSSION

Breast MRI provides excellent sensitivity for lesion detection; however, distinguishing benign from malignant lesions remains challenging because benign entities may demonstrate enhancement patterns overlapping with malignancies. This limitation is particularly evident in BI-RADS 4 lesions, where the probability of malignancy varies widely, and additional biopsies are frequently conducted despite benign outcomes. Recent studies have therefore investigated radiomics and ML approaches as quantitative decision-support tools capable of extracting imaging biomarkers beyond conventional visual interpretation.

In this study, we developed and evaluated a patient-aware ML framework for differentiating benign and malignant breast lesions using multiparametric MRI radiomics. Unlike previous investigations that mainly focused on individual MRI sequences or limited combinations of imaging biomarkers, this work comprehensively assessed multiple single- and multi-sequence radiomic configurations while maintaining strict patient-level separation during model development. This approach addresses an important methodological challenge in radiomics studies, where small sample sizes, high-dimensional feature spaces, and potential data leakage can lead to overly optimistic estimates of model performance.

A major observation of this study was that increasing the number of imaging sources did not consistently improve classification performance. Although multiparametric MRI provides complementary biological information, combining additional sequences also substantially increases feature dimensionality and the risk of incorporating redundant or unstable predictors. Therefore, successful multiparametric radiomics depends on identifying complementary information rather than maximizing the number of input sequences.

Ellmann et al. [23] demonstrated that ML applied to MRI-derived imaging parameters could provide objective decision rules for suspicious breast masses and potentially reduce unnecessary biopsies. Their study emphasized the potential role of AI as a complementary tool for improving MRI-based decision-making rather than replacing radiologist assessment.

Previous studies have reported heterogeneous results regarding the benefit of multiparametric MRI fusion. Naranjo et al. [24] investigated combined DCE and DWI radiomics and demonstrated that multiparametric models could improve lesion classification compared with individual MRI modalities, supporting the complementary role of contrast enhancement and diffusion information. Their work showed that DCE-derived models captured enhancement-related characteristics, while DWI contributed additional information related to tissue diffusivity. Similarly, Naranjo et al. [25] later compared multiparametric radiomic models with radiologist interpretation and demonstrated that integration of radiomics with conventional imaging biomarkers could enhance diagnostic performance. However, other investigations have shown that additional sequences do not always provide meaningful improvement. Zhang et al. [26] evaluated MRI radiomics for lymphovascular invasion prediction and reported that although multiple MRI sequences were available, careful feature selection was required because additional imaging information did not automatically translate into better predictive performance. These findings support our observation that optimized feature fusion is more important than indiscriminate accumulation of imaging sources.

The importance of feature-selection strategy is particularly relevant in small radiomic datasets. Radiomics generates hundreds of quantitative descriptors describing intensity distribution, texture, and morphology, while

clinical datasets frequently contain substantially fewer patients [27]. Without appropriate dimensionality reduction, models are vulnerable to overfitting and limited generalizability. Previous studies have addressed this challenge using different approaches, including LASSO, correlation-based filtering, mutual information methods, and embedded machine-learning selection strategies. Zhao et al. [18] demonstrated that combining feature-selection approaches with machine learning classifiers enabled successful discrimination of benign and malignant breast lesions using MRI texture features. Their study evaluated multiple feature-selection methods, including distance correlation, gradient boosting-based selection, LASSO, random forest, and XGBoost-based approaches, highlighting the importance of systematic feature reduction before classification.

In the current study, the frequent selection of Elastic Net followed by mRMR reflects the complementary advantages of embedded regularization and relevance–redundancy optimization. Elastic Net reduces the contribution of weak or highly correlated predictors, whereas mRMR prioritizes informative features while minimizing redundancy. This combination is particularly appropriate for multiparametric MRI radiomics, where many extracted features describe related biological phenomena.

The three most influential features identified by the explainability analysis further supported the biological relevance of the developed model. The ADC-derived first-order intensity features, particularly ADC intensity histogram median and median intensity, were the most important predictors. These parameters represent the distribution of water diffusivity within the lesion and may reflect differences in cellular density and extracellular space. Malignant breast lesions typically demonstrate restricted diffusion due to increased cellularity, resulting in lower ADC values compared with benign lesions. Therefore, the contribution of ADC intensity features is consistent with the known association between diffusion characteristics and tumor cellular architecture.

The third most important feature was derived from Post1 DCE-T1W imaging and represented enhancement intensity distribution. First-order DCE features quantify the heterogeneity and magnitude of contrast uptake within lesions, which are influenced by tumor vascularity, permeability, and extracellular matrix characteristics. Malignant lesions often exhibit heterogeneous enhancement due to abnormal angiogenesis and variable perfusion patterns, making DCE-derived intensity features valuable biomarkers for lesion characterization.

In addition, the contribution of Post1 morphological features, including compactness and sphericity, highlights the importance of lesion architecture in malignancy assessment. Irregular growth patterns and invasive tumor expansion may result in less organized lesion boundaries compared with benign lesions. The selected radiomic characteristics in this study were also biologically consistent with previous literature [17-19]. Sun et al. [19] showed that radiomic analysis of diffusion-weighted MRI provided superior discrimination compared with conventional mean diffusion measurements, indicating that quantitative heterogeneity captured by radiomics contains additional biological information beyond simple ADC values. Their findings also highlighted the importance of first-order intensity features, supporting the relevance of intensity-distribution descriptors in breast lesion characterization.

DCE-MRI radiomics provides complementary information related to vascularity, permeability, and enhancement heterogeneity. D’Amico et al. [28] demonstrated that radiomic analysis of dynamic contrast-enhanced MRI could differentiate malignant from benign enhancing breast foci using machine-learning approaches, showing that quantitative enhancement patterns contain clinically relevant information beyond visual assessment. Similarly, Ji

et al. [29] developed and independently evaluated a machine-learning system based on DCE-MRI radiomic features for breast lesion diagnosis, demonstrating the feasibility of radiomic signatures as computer-aided diagnostic tools.

Morphological and texture features have also been repeatedly associated with malignant biological behavior [29, 30]. Matsuda et al. [17] demonstrated that radiomic models incorporating lesion shape, texture, and clinical information could improve breast lesion classification, and that combining ML with conventional BI-RADS assessment improved diagnostic performance. These findings support the interpretation that radiomic descriptors may indirectly represent underlying tumor characteristics, including irregular growth patterns, heterogeneity, and altered vascular structure.

The variation in optimal classifier selection across studies indicates that no single machine-learning algorithm is universally superior for radiomic classification. Previous breast MRI studies have reported successful applications of SVM, random forest, KNN, logistic regression, and boosting-based algorithms [31, 32]. D'Amico et al. [28] identified KNN as the best-performing classifier for their selected radiomic signature, demonstrating the effectiveness of nonlinear classification in small datasets. In contrast, Ellmann et al. [23] achieved strong performance using an SVM-based model developed from quantitative MRI parameters and clinical information. These differences emphasize that classifier performance depends strongly on the characteristics of the feature space, sample size, and modelling strategy rather than the algorithm alone.

The limited contribution of SMOTE in our study further demonstrates the importance of evaluating imbalance correction strategies rather than applying them automatically. Although SMOTE improved performance in some configurations, the overall benefit was minimal, and the best-performing model was obtained without oversampling. This may be related to the relatively balanced distribution of benign and malignant lesions in our dataset. Previous ML studies have highlighted that class imbalance can bias models toward the majority class and that oversampling methods should be carefully evaluated [11, 33-35]. However, oversampling may also introduce synthetic patterns that do not represent true biological variation, particularly in small radiomic datasets.

An important methodological strength of the present study is the strict patient-aware validation strategy. Radiomic datasets frequently contain multiple lesions or regions of interest from the same patient, creating a risk of information leakage if samples from the same patient are distributed between training and testing subsets. Such leakage may result in artificially inflated performance estimates because models can learn patient-specific imaging characteristics rather than lesion-specific biomarkers. By maintaining patient-level separation throughout feature selection, optimization, and evaluation, this study provides a more realistic estimation of model generalizability.

Despite promising results, several limitations should be considered. First, this was a retrospective single-center study with a relatively limited cohort size, which may restrict external generalizability. Second, radiomic features are sensitive to variations in acquisition parameters, reconstruction methods, preprocessing strategies, and segmentation accuracy. Third, the current framework incorporated imaging-derived biomarkers only and did not include clinical variables, molecular subtype information, or BI-RADS descriptors that may provide additional complementary information. Future studies should investigate multicenter external validation, harmonized imaging protocols, radiomic feature stability analysis, and integrated clinical-radiomic models.

In conclusion, this study demonstrates that reliable breast MRI radiomic classification requires careful optimization of imaging inputs, feature-selection strategies, and validation methodology. The findings support the concept that biologically meaningful and methodologically robust feature fusion is more valuable than simply increasing the number of imaging sequences. Patient-aware ML frameworks incorporating multiparametric MRI radiomics may provide a reproducible quantitative tool to assist breast lesion characterization and support future personalized diagnostic pathways.

5. CONCLUSIONS

This study developed and evaluated a patient-aware explainable ML framework for classification of benign and suspicious breast lesions based on BI-RADS assessment using multiparametric MRI radiomics. Systematic comparison of single- and multi-sequence radiomic configurations demonstrated that optimal performance depends on selective integration of complementary MRI biomarkers rather than simply increasing the number of imaging sources.

The proposed workflow incorporated rigorous strategies to improve model reliability, including patient-level data separation, nested CV, feature-selection optimization, and explainability analysis. The identified radiomic predictors reflected biologically meaningful characteristics related to tissue diffusivity, contrast enhancement behavior, and lesion morphology, supporting the interpretability of the developed model.

Although the results demonstrate the feasibility of MRI radiomics for non-invasive breast lesion characterization, further validation in larger multicenter cohorts is required before clinical translation. Future studies integrating clinical, pathological, and molecular information may further improve the robustness and clinical utility of radiomics-based decision-support systems.

6. LIST OF ABBREVIATIONS

ADC; Apparent Diffusion Coefficient; **AI**; Artificial Intelligence; **AUC**; Area Under the Receiver Operating Characteristic Curve; **BI-RADS**; Breast Imaging Reporting and Data System; **CV**; Cross-Validation; **DCA**; Decision Curve Analysis; **DCE**; Dynamic Contrast-Enhanced; **DWI**; Diffusion-Weighted Imaging; **EN**; Elastic Net; **EPI**; Echo-Planar Imaging; **GLCM**; Gray-Level Co-occurrence Matrix; **GLRLM**; Gray-Level Run-Length Matrix; **GLSZM**; Gray-Level Size Zone Matrix; **GRE**; Gradient Echo; **ITK-SNAP**; Insight Toolkit-SNAP; **KNN**; K-Nearest Neighbors; **LightGBM**; Light Gradient Boosting Machine; **mRMR**; Minimum Redundancy Maximum Relevance; **ML**; Machine Learning; **MRI**; Magnetic Resonance Imaging; **NGLDM**; Neighborhood Gray-Level Difference Matrix; **Post1**; First Post-Contrast Phase; **Post2**; Second Post-Contrast Phase; **Pre**; Pre-Contrast Phase; **PR-AUC**; Precision–Recall Area Under the Curve; **RBF**; Radial Basis Function; **RF**; Random Forest; **ROI**; Region of Interest; **ROC**; Receiver Operating Characteristic; **SD**; Standard Deviation; **SHAP**; SHapley Additive exPlanations; **SMOTE**; Synthetic Minority Oversampling Technique; **SPAIR**; Spectral Attenuated Inversion Recovery; **STIR**; Short Tau Inversion Recovery; **SVM**; Support Vector Machine; **T1W**; T1-Weighted Imaging; **T2W**; T2-Weighted Imaging; **TE**; Echo Time; **TI**; Inversion Time; **TPE**; Tree-structured

Parzen Estimator; **TR**; Repetition Time; **TSE**; Turbo Spin Echo; **VIBE**; Volumetric Interpolated Breath-hold Examination; **XAI**; Explainable Artificial Intelligence; **XGBoost**; Extreme Gradient Boosting.

DECLARATIONS

Ethics approval and consent to participate

This retrospective study was approved by the Ethics Committee of Iran University of Medical Sciences (**IR.IUMS.FMD.REC.1405.703**). All methods were conducted in accordance with the Declaration of Helsinki (as revised in 2013) and relevant institutional guidelines and regulations. Due to the retrospective nature of the study and the use of anonymized imaging data obtained as part of routine clinical care, the Ethics Committee waived the requirement for informed consent. Patient data were anonymized and de-identified prior to analysis, ensuring confidentiality and compliance with data protection standards.

Consent for publication

Not applicable.

Availability of data and materials

The datasets used and analyzed during the current study will be available from the corresponding author upon reasonable request.

Competing interests

The authors declare no competing interests.

Funding

This research received no external funding.

Authors' contributions

Study conception and design: MB, EY, SRK; Study supervision and conceptual feedback: MB, SRK, EY; Data collection and preprocessing: MB, EY, ZF, ZB, HN, EM; Image interpretation and clinical oversight: HN, EM, PF, SHT, HA; Machine learning model development and analysis: EY, ME, SRK; Result interpretation: EY, MB, SRK, MS; Visualization: EY, ME, ZF; Manuscript writing: EY, ME, ZF; All authors read, edited, and approved the final manuscript.

Acknowledgments

This work was supported by the School of Medicine, Iran University of Medical Sciences (IUMS), Grant 37857.

REFERENCES

1. Siegel RL, Miller KD, Jemal A. Cancer statistics, 2020. CA: a cancer journal for clinicians. 2020;70(1):7-30.

2. Burnside MES, Sickles MEA, Bassett MLW, Lee MCH, Ikeda MDM, Mendelson MEB, et al. The ACR BI-RADS® experience: learning from history. *Journal of the American College of Radiology*. 2009;6(12):851-60.
3. Bennani-Baiti B, Bennani-Baiti N, Baltzer PA. Diagnostic performance of breast magnetic resonance imaging in non-calcified equivocal breast findings: results from a systematic review and meta-analysis. *PloS one*. 2016;11(8):e0160346.
4. Bennani-Baiti B, Baltzer PA. MR imaging for diagnosis of malignancy in mammographic microcalcifications: a systematic review and meta-analysis. *Radiology*. 2017;283(3):692-701.
5. Mann RM, Kuhl CK, Kinkel K, Boetes C. Breast MRI: guidelines from the European society of breast imaging. *European radiology*. 2008;18(7):1307-18.
6. Grimm LJ, Anderson AL, Baker JA, Johnson KS, Walsh R, Yoon SC, et al. Interobserver variability between breast imagers using the fifth edition of the BI-RADS MRI lexicon. *American Journal of Roentgenology*. 2015;204(5):1120-4.
7. Reig B, Heacock L, Geras KJ, Moy L. Machine learning in breast MRI. *Journal of magnetic resonance imaging*. 2020;52(4):998-1018.
8. Tahmassebi A, Wengert GJ, Helbich TH, Bago-Horvath Z, Alaei S, Bartsch R, et al. Impact of machine learning with multiparametric magnetic resonance imaging of the breast for early prediction of response to neoadjuvant chemotherapy and survival outcomes in breast cancer patients. *Investigative radiology*. 2019;54(2):110.
9. Doan LMT, Shahhosseini K, Verma S, Marefat A, Locicero G, Verma S, et al. Bridging modalities with AI: a review of AI advances in multimodal biomedical imaging. *Communications Engineering*. 2026;5(1):30.
10. Ismail FA, Karim MKA, Kechik MMA, Kamal I, Ismail Z, Salim EM, et al. Radiomics-based Machine Learning Models for Classifying Breast Cancer on Dynamic-Contrast Enhanced MRI through Multi-Observer Analysis. *Current Medical Imaging*. 2026;22(1):E15734056403426.
11. Mahesh T, Geman O, Margala M, Guduri M. The stratified K-folds cross-validation and class-balancing methods with high-performance ensemble classifiers for breast cancer classification. *Healthcare Analytics*. 2023;4:100247.
12. Radak M, Lafta HY, Fallahi H. Machine learning and deep learning techniques for breast cancer diagnosis and classification: a comprehensive review of medical imaging studies. *Journal of Cancer Research and Clinical Oncology*. 2023;149(12):10473-91.
13. Lambin P, Leijenaar RT, Deist TM, Peerlings J, De Jong EE, Van Timmeren J, et al. Radiomics: the bridge between medical imaging and personalized medicine. *Nature reviews Clinical oncology*. 2017;14(12):749-62.
14. Kumar V, Gu Y, Basu S, Berglund A, Eschrich SA, Schabath MB, et al. Radiomics: the process and the challenges. *Magnetic resonance imaging*. 2012;30(9):1234-48.
15. Gillies RJ, Kinahan PE, Hricak H. Radiomics: images are more than pictures, they are data. *Radiology*. 2016;278(2):563.
16. Qi Y-J, Su G-H, You C, Zhang X, Xiao Y, Jiang Y-Z, et al. Radiomics in breast cancer: Current advances and future directions. *Cell Reports Medicine*. 2024;5(9).
17. Matsuda T, Matsuda M, Haque H, Fuchibe S, Matsumoto M, Shiraishi Y, et al. Diagnostic accuracy of a machine learning model using radiomics features from breast synthetic MRI. *BMC Medical Imaging*. 2025;25(1):399.
18. Zhao Y, Chen R, Zhang T, Chen C, Muhelisa M, Huang J, et al. MRI-based machine learning in differentiation between benign and malignant breast lesions. *Frontiers in Oncology*. 2021;11:552634.
19. Sun K, Jiao Z, Zhu H, Chai W, Yan X, Fu C, et al. Radiomics-based machine learning analysis and characterization of breast lesions with multiparametric diffusion-weighted MR. *Journal of translational medicine*. 2021;19(1):443.
20. Yushkevich PA, Piven J, Hazlett HC, Smith RG, Ho S, Gee JC, et al. User-guided 3D active contour segmentation of anatomical structures: Significantly improved efficiency and reliability. *NeuroImage*. 2006;31(3):1116-28.
21. Nioche C, Orlhac F, Boughdad S, Reuzé S, Goya-Outi J, Robert C, et al. LIFEx: a freeware for radiomic feature calculation in multimodality imaging to accelerate advances in the characterization of tumor heterogeneity. *Cancer research*. 2018;78(16):4786-9.
22. Hatt M, Vallieres M, Visvikis D, Zwanenburg A. IBSI: an international community radiomics standardization initiative. *Soc Nuclear Med*; 2018.
23. Ellmann S, Wenkel E, Dietzel M, Bielowski C, Vesal S, Maier A, et al. Implementation of machine learning into clinical breast MRI: Potential for objective and accurate decision-making in suspicious breast masses. *Plos one*. 2020;15(1):e0228446.

24. Daimiel Naranjo I, Gibbs P, Reiner JS, Lo Gullo R, Sooknanan C, Thakur SB, et al. Radiomics and machine learning with multiparametric breast MRI for improved diagnostic accuracy in breast cancer diagnosis. *Diagnostics*. 2021;11(6):919.
25. Daimiel Naranjo I, Gibbs P, Reiner JS, Lo Gullo R, Thakur SB, Jochelson MS, et al. Breast lesion classification with multiparametric breast MRI using radiomics and machine learning: a comparison with radiologists' performance. *Cancers*. 2022;14(7):1743.
26. Zhang C, Zhong M, Liang Z, Zhou J, Wang K, Bu J. MRI-based radiomic and machine learning for prediction of lymphovascular invasion status in breast cancer. *BMC Medical Imaging*. 2024;24(1):322.
27. Soltani S, Aliasgharzadeh AA, Fadavi P, Bagherpour Z, Moradi H, Safari M, et al. Enhanced prediction of radiation-induced skin toxicity in breast cancer patients using a hybrid dosiomics-clinical model. *Journal of Radiation Research and Applied Sciences*. 2025;18(2):101382.
28. D'Amico NC, Grossi E, Valbusa G, Rigioli F, Colombo B, Buscema M, et al. A machine learning approach for differentiating malignant from benign enhancing foci on breast MRI. *European Radiology Experimental*. 2020;4(1):5.
29. Ji Y, Li H, Edwards AV, Papaioannou J, Ma W, Liu P, et al. Independent validation of machine learning in diagnosing breast Cancer on magnetic resonance imaging within a single institution. *Cancer Imaging*. 2019;19(1):64.
30. Gholizade M, Yazdani E, Hosseini-Baharanchi FS, Nikoofar A, Esmaili G, Goli-Ahmadabad F, et al. Variations in radiomic features of the femoral head and neck during helical tomotherapy in prostate and rectal cancer patients. *BMC cancer*. 2025;25(1):1509.
31. Mahmoudi A, Sadeghi AZ, Iraj H, Barahman M, Saadatmand P, Yazdani E, et al. Machine learning-based integration of dosiomics and pre-radiotherapy multimodal MRI radiomics for survival stratification in patients with glioblastoma multiforme. *Radiation Oncology*. 2025;20(1):187.
32. Bagherpour Z, Safari M, Fadavi P, Haghpanah M, Beigi M. Predicting Radiation-Induced Skin Toxicity in Breast Cancer: A Machine Learning Approach Combining Radiomic and Dosimetric Features. *Journal of Medical and Biological Engineering*. 2025;45(2):211-22.
33. Gao X, Xie D, Zhang Y, Wang Z, Chen C, He C, et al. A comprehensive survey on imbalanced data learning. *Frontiers of Computer Science*. 2026;20(11):2011622.
34. Nikpour B, Rahmati F, Mirzaei B, Nezamabadi-pour H. A comprehensive review on data-level methods for imbalanced data classification. *Expert Systems with Applications*. 2026;295:128920.
35. Mahmoudi A, Sadeghi AZ, Iraj H, Barahman M, Yazdani E, Mahdavi SR. Correlation of contrast-enhancing tumor region on pre-radiotherapy MRI with overall survival in glioblastoma. *European Journal of Medical Research*. 2025;30(1):1121.